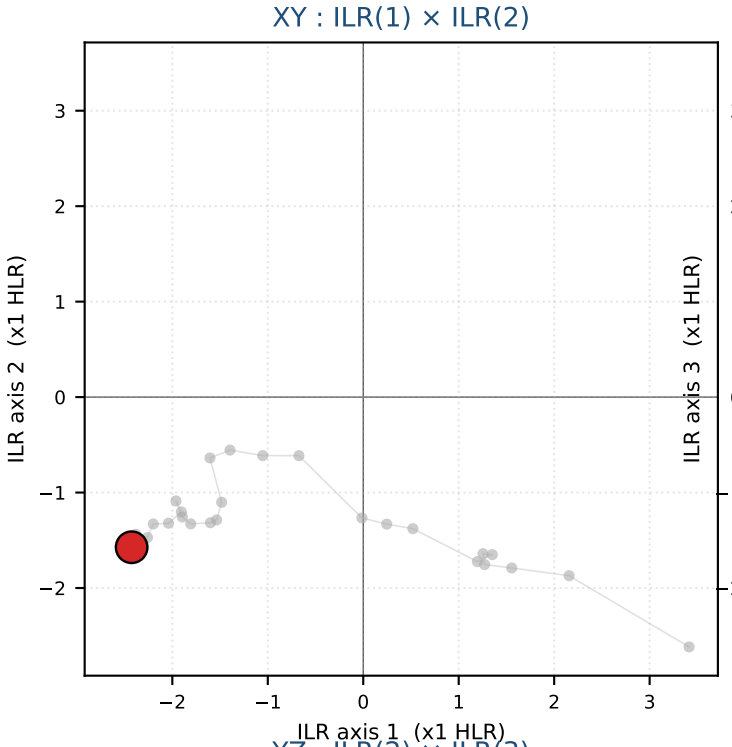
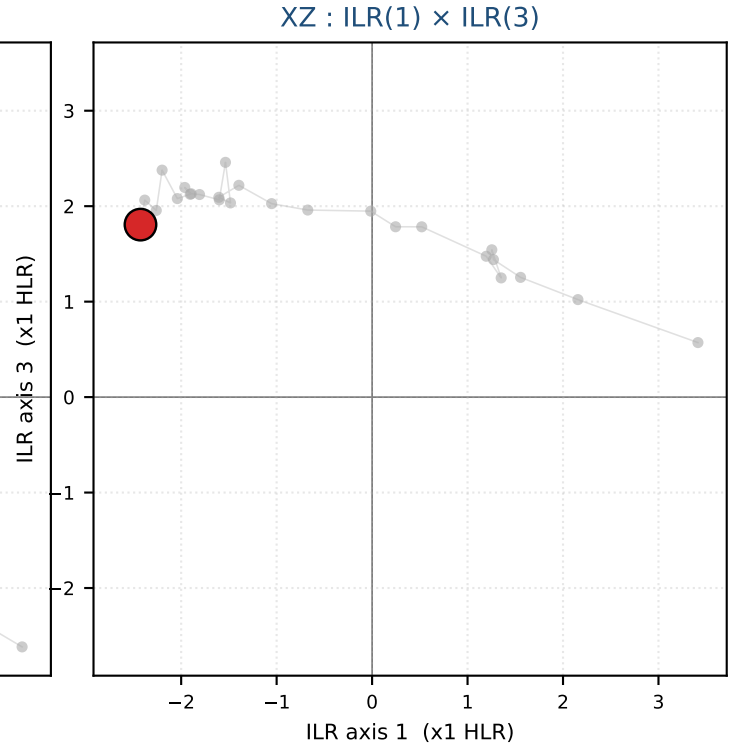
STAGE 1 v4 | t=0 | 2000 | D=9 T=26 | level 1 first-principles

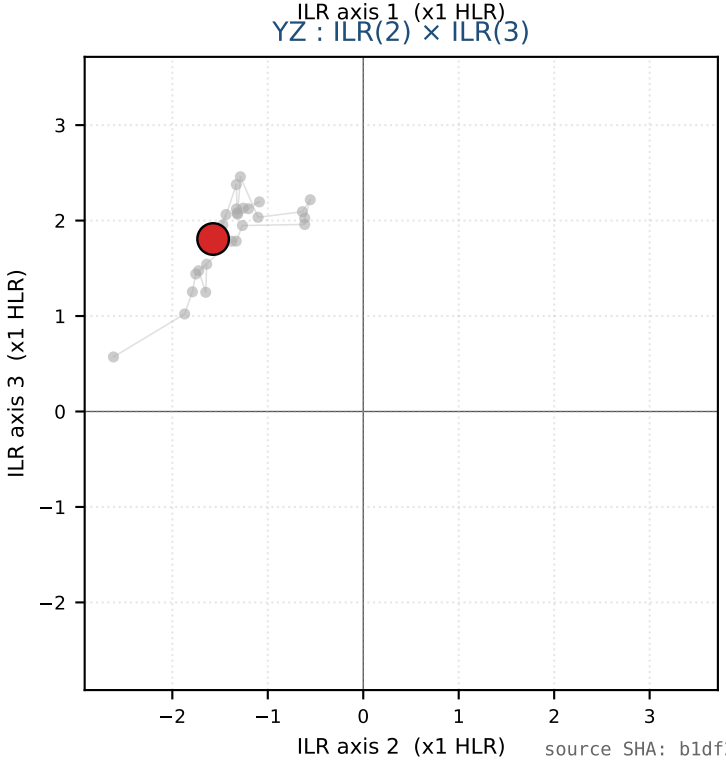
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

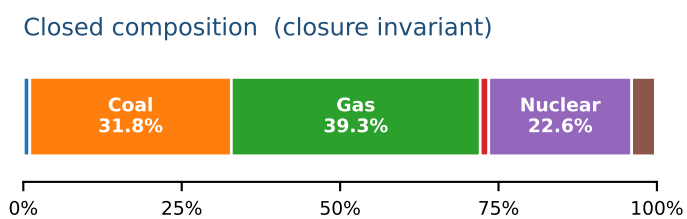




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0103	+6.832
Coal	0.3181	+10.263
Gas	0.3927	+10.474
Hydro	0.0135	+7.103
Nuclear	0.2256	+9.919
Other Fossil	0.0373	+8.119
Other Renewabl	0.0000	-29.063
Solar	0.0000	-29.063
Wind	0.0025	+5.414





closure: Σ = 1.000000 OK

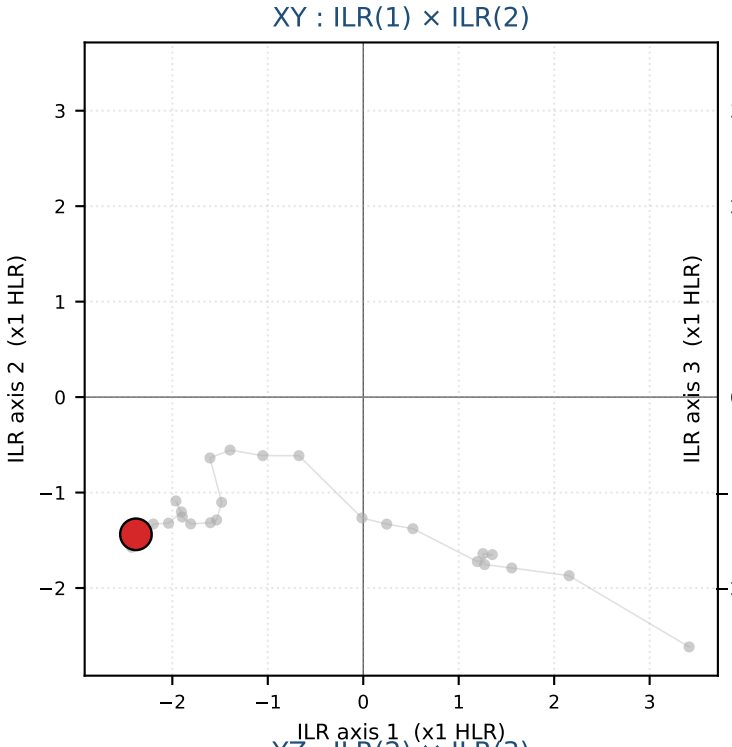
display factor: x1 (raw HLR x 1)

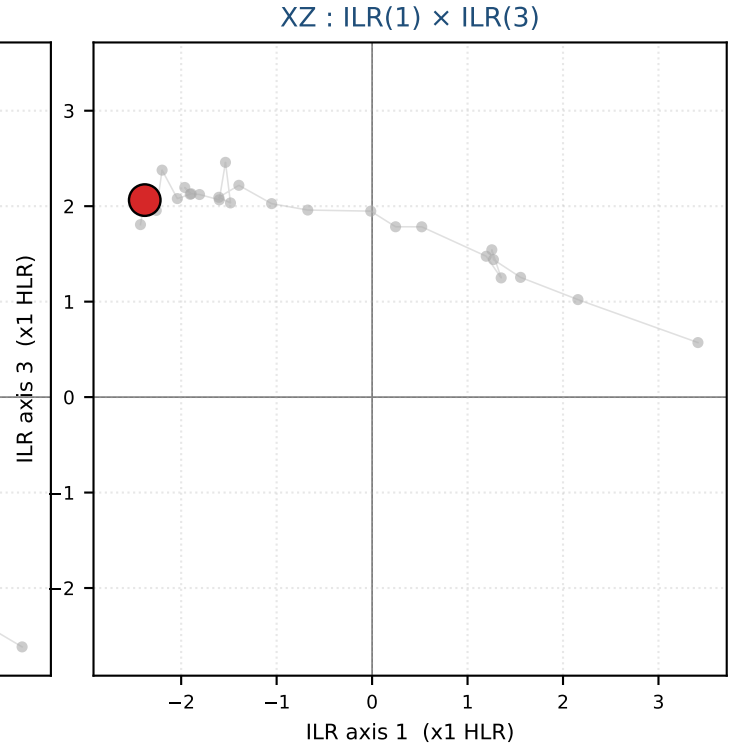
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2001 | D=9 T=26 | level 1 first-principles

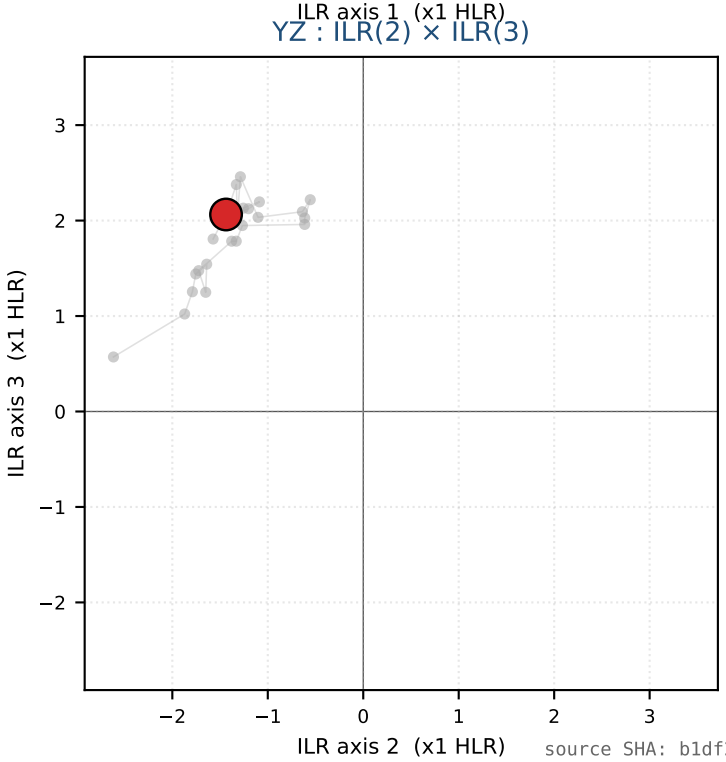
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

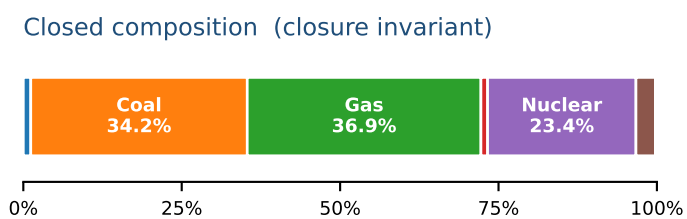




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0118	+7.001
Coal	0.3417	+10.369
Gas	0.3688	+10.445
Hydro	0.0105	+6.889
Nuclear	0.2341	+9.991
Other Fossil	0.0306	+7.955
Other Renewabl	0.0000	-29.049
Solar	0.0000	-29.049
Wind	0.0025	+5.449





closure: Σ = 1.000000 OK

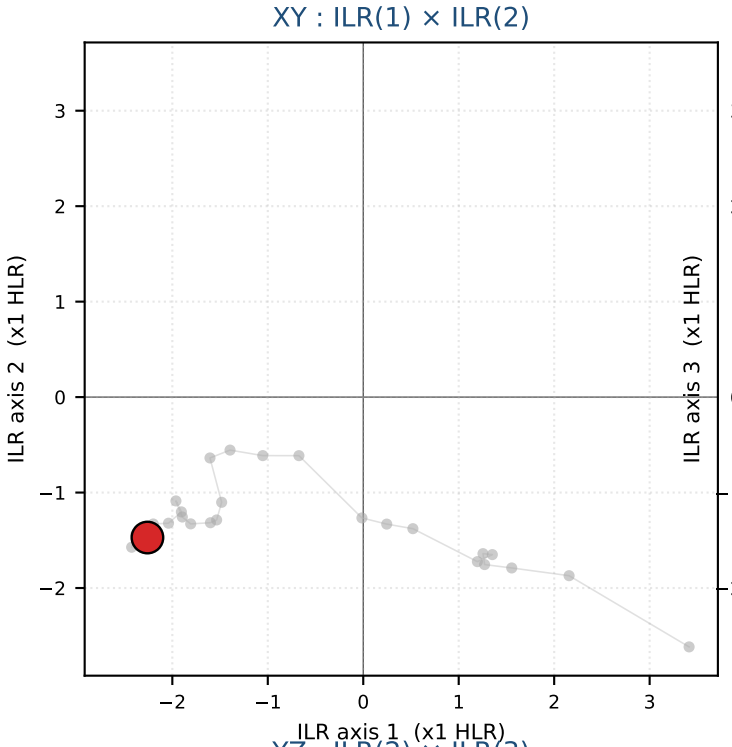
display factor: x1 (raw HLR x 1)

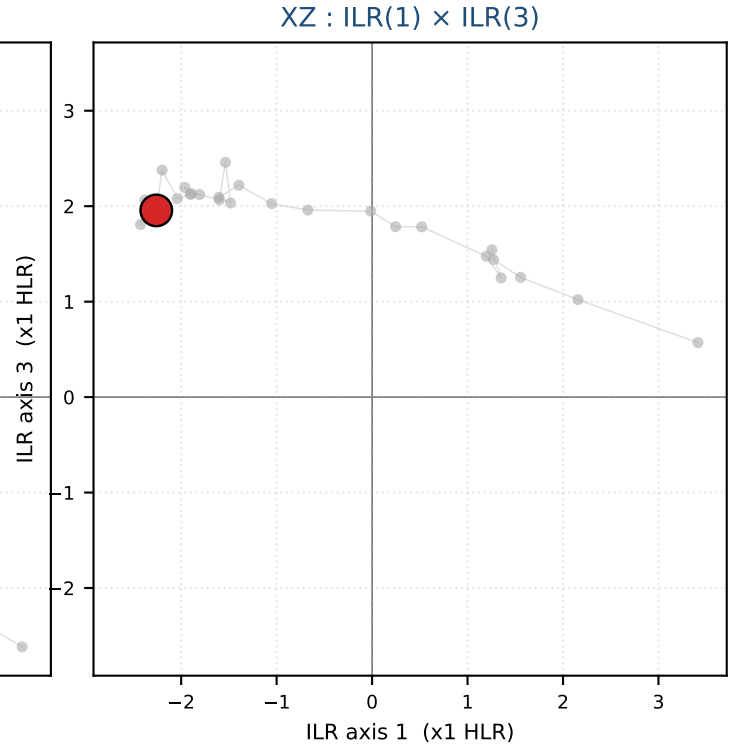
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2002 | D=9 T=26 | level 1 first-principles

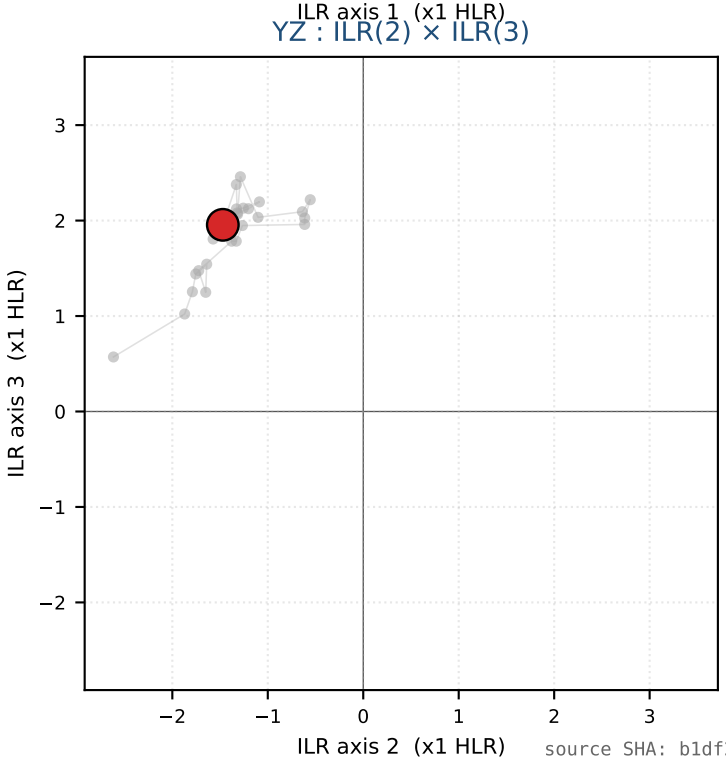
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

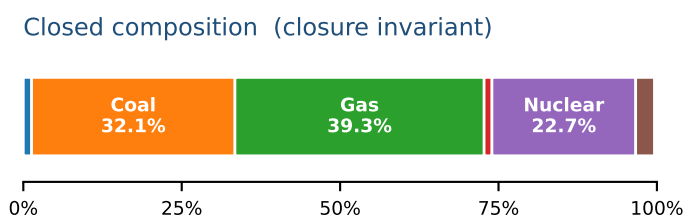




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0131	+7.056
Coal	0.3209	+10.253
Gas	0.3932	+10.457
Hydro	0.0124	+6.997
Nuclear	0.2269	+9.907
Other Fossil	0.0302	+7.891
Other Renewabl	0.0000	-29.108
Solar	0.0000	-29.108
Wind	0.0032	+5.654





closure: Σ = 1.000000 OK

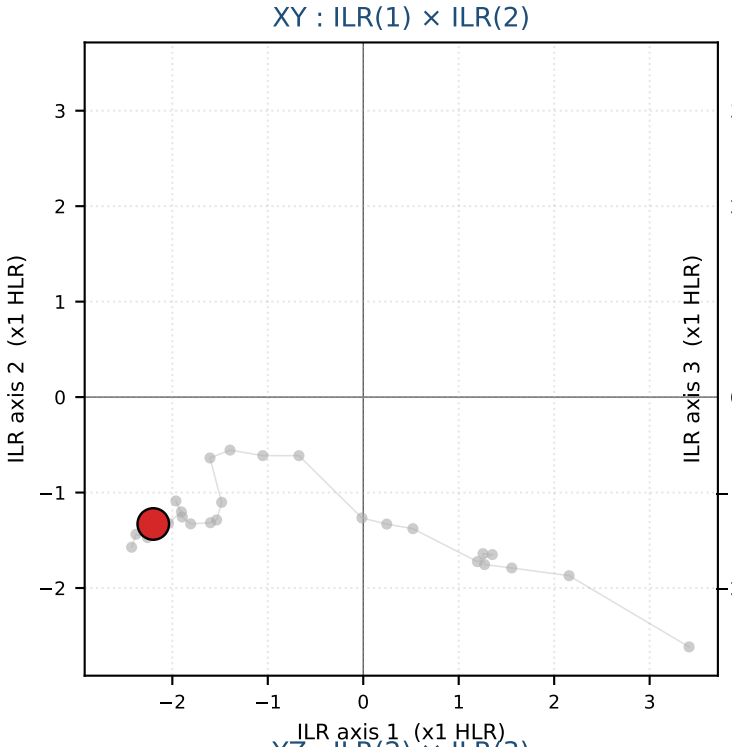
display factor: x1 (raw HLR x 1)

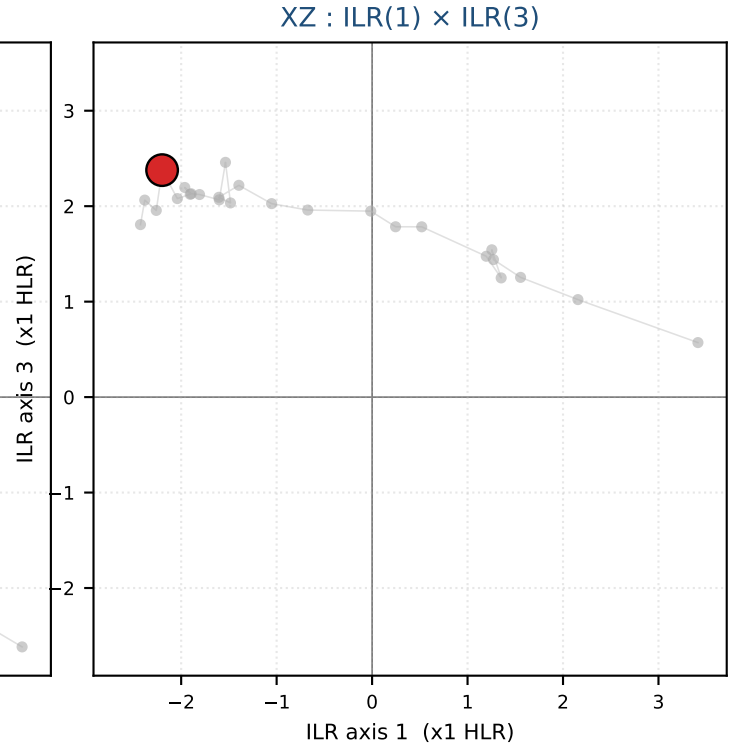
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2003 | D=9 T=26 | level 1 first-principles

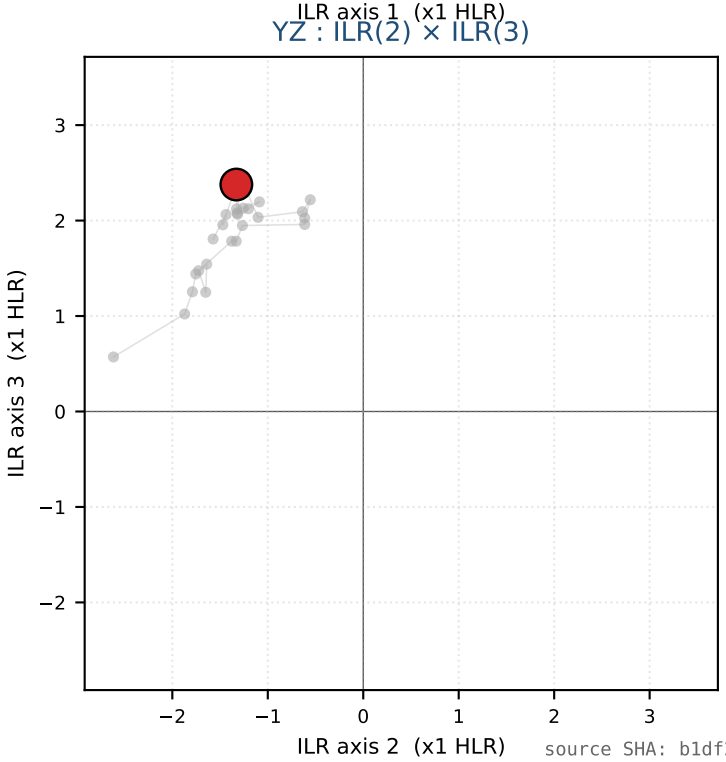
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

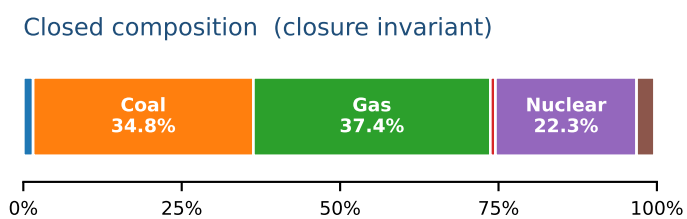




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0155	+7.261
Coal	0.3477	+10.372
Gas	0.3739	+10.444
Hydro	0.0081	+6.614
Nuclear	0.2227	+9.926
Other Fossil	0.0289	+7.883
Other Renewabl	0.0000	-29.098
Solar	0.0000	-29.098
Wind	0.0032	+5.696





closure: Σ = 1.000000 OK

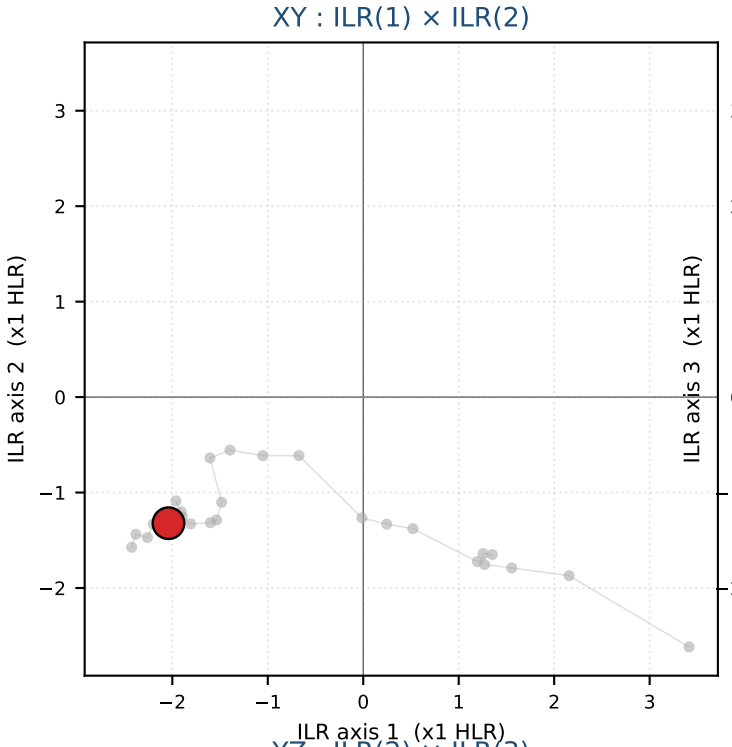
display factor: x1 (raw HLR x 1)

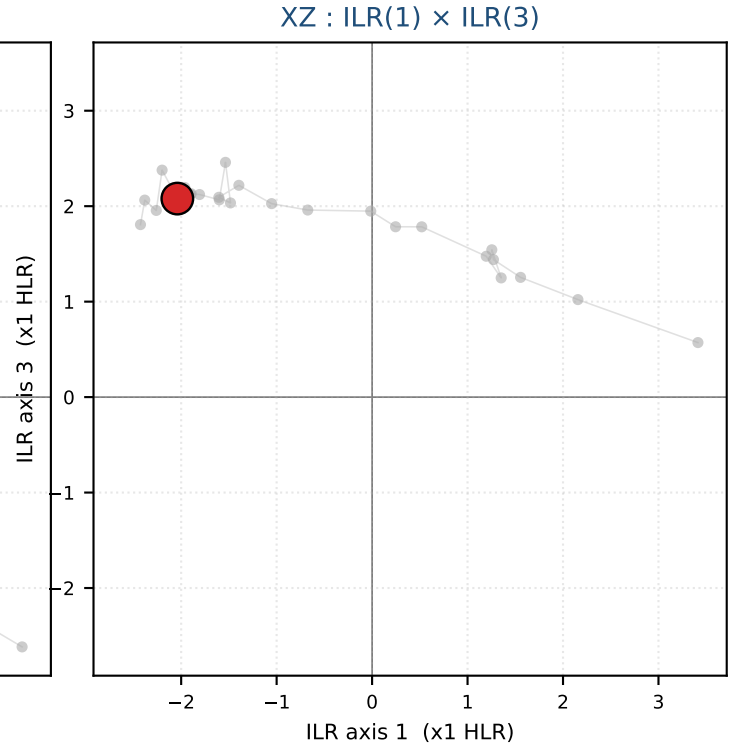
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2004 | D=9 T=26 | level 1 first-principles

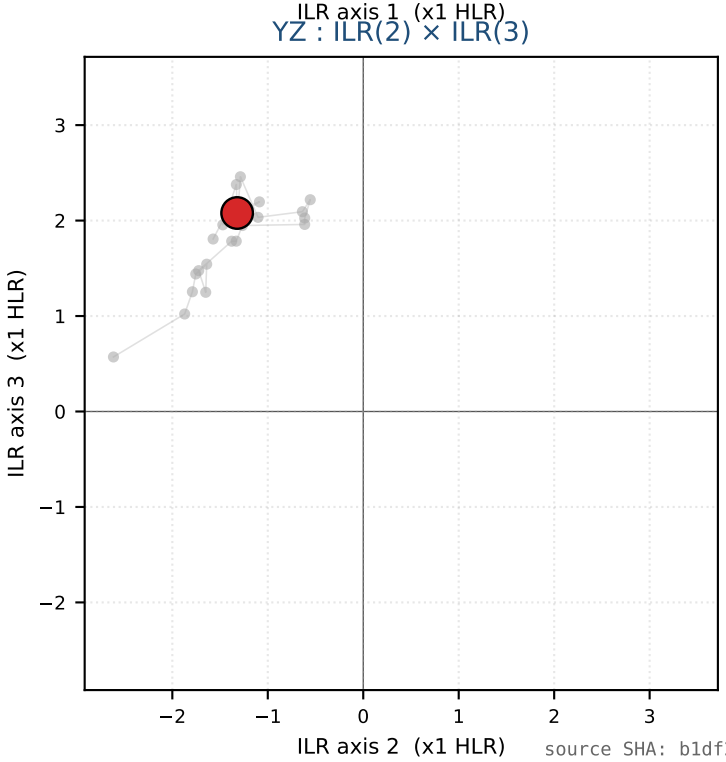
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

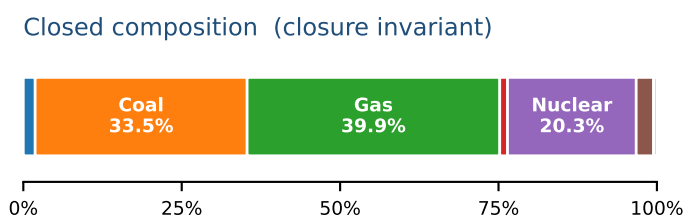




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0187	+7.344
Coal	0.3346	+10.229
Gas	0.3987	+10.404
Hydro	0.0123	+6.925
Nuclear	0.2031	+9.730
Other Fossil	0.0277	+7.739
Other Renewabl	0.0000	-29.191
Solar	0.0000	-29.191
Wind	0.0049	+6.011





closure: Σ = 1.000000 OK

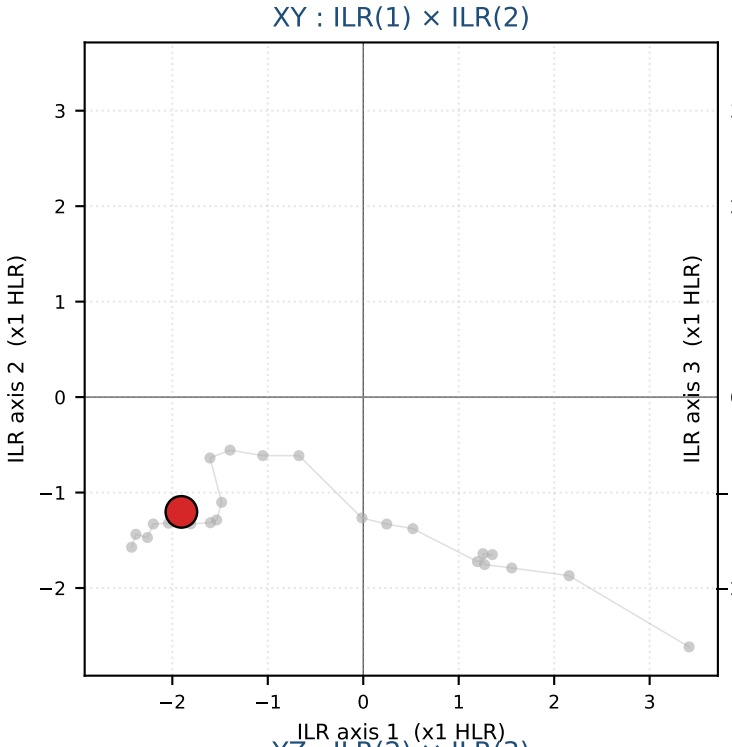
display factor: x1 (raw HLR x 1)

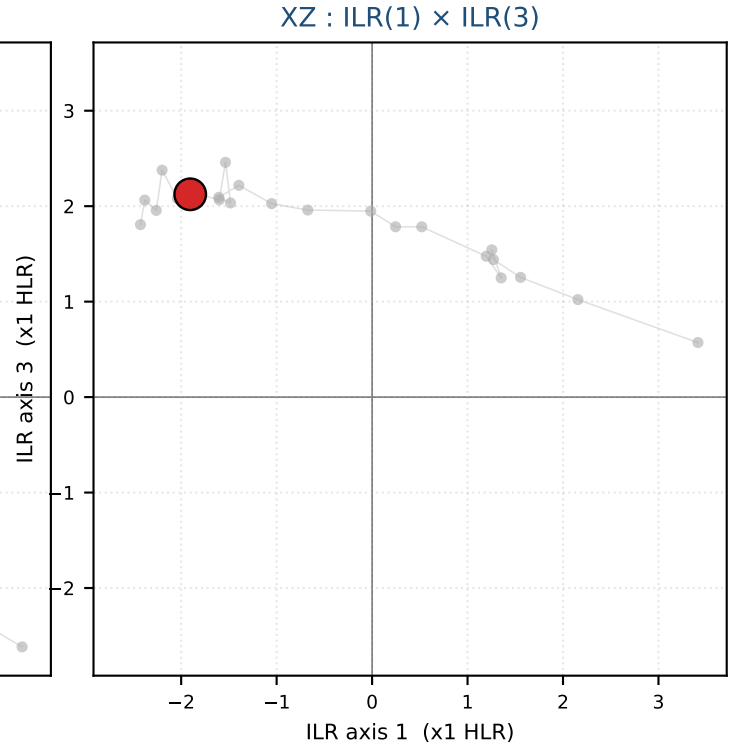
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2005 | D=9 T=26 | level 1 first-principles

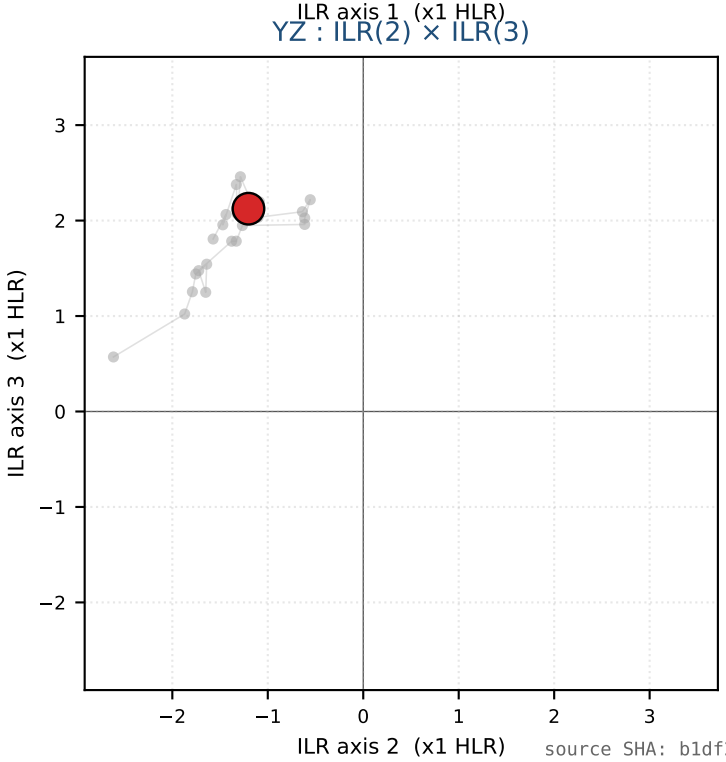
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

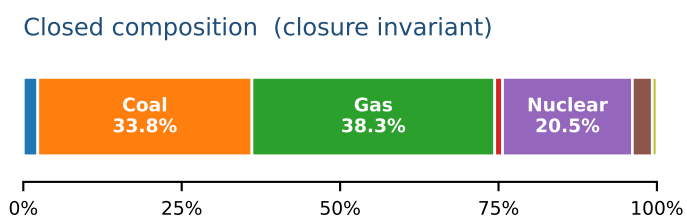




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0228	+4.144
Coal	0.3380	+6.838
Gas	0.3832	+6.963
Hydro	0.0124	+3.529
Nuclear	0.2049	+6.337
Other Fossil	0.0314	+4.463
Other Renewabl	0.0000	-32.603
Solar	0.0000	-2.670
Wind	0.0073	+3.000





closure: Σ = 1.000000 OK

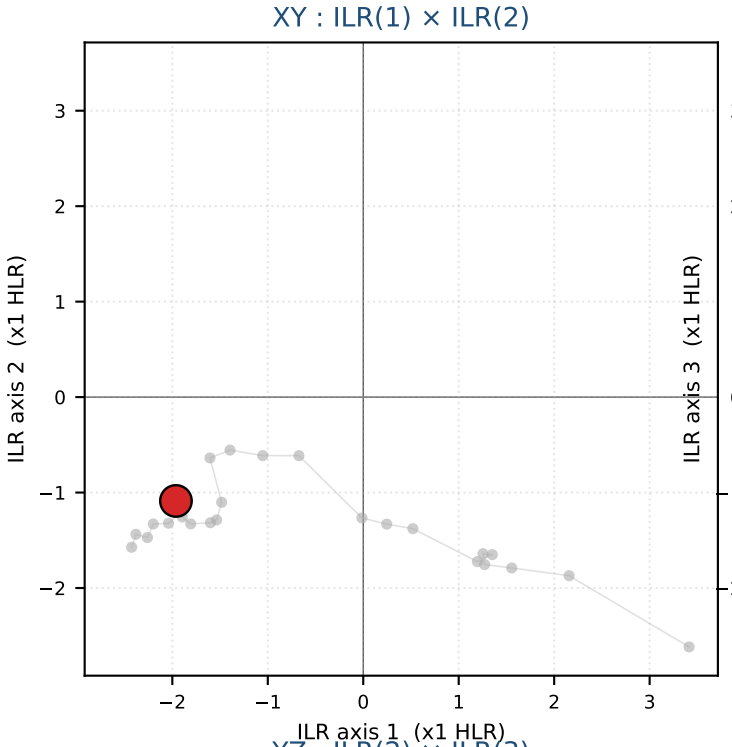
display factor: x1 (raw HLR x 1)

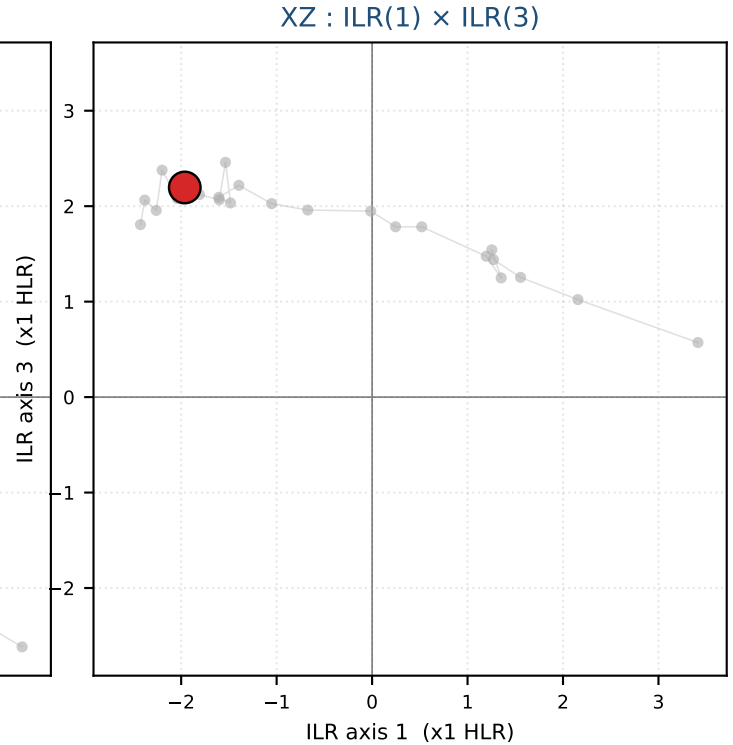
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2006 | D=9 T=26 | level 1 first-principles

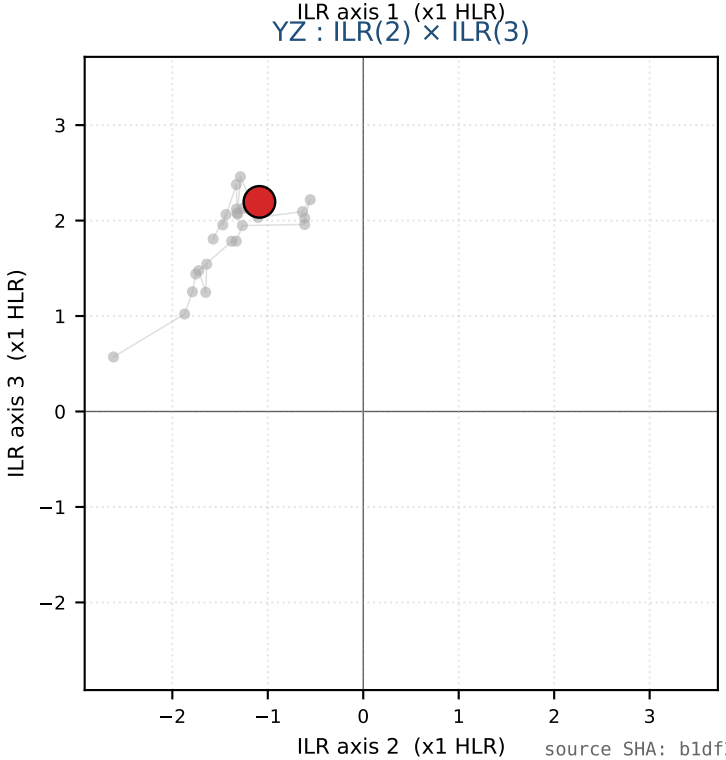
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

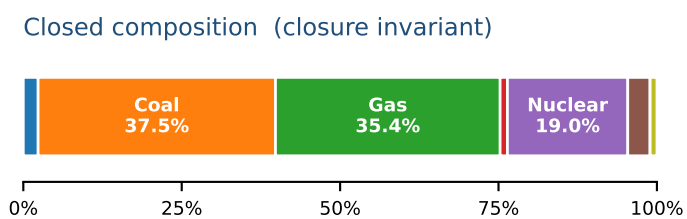




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0234	+4.121
Coal	0.3747	+6.896
Gas	0.3545	+6.840
Hydro	0.0116	+3.417
Nuclear	0.1899	+6.216
Other Fossil	0.0353	+4.535
Other Renewabl	0.0000	-32.646
Solar	0.0000	-2.712
Wind	0.0106	+3.333





closure: Σ = 1.000000 OK

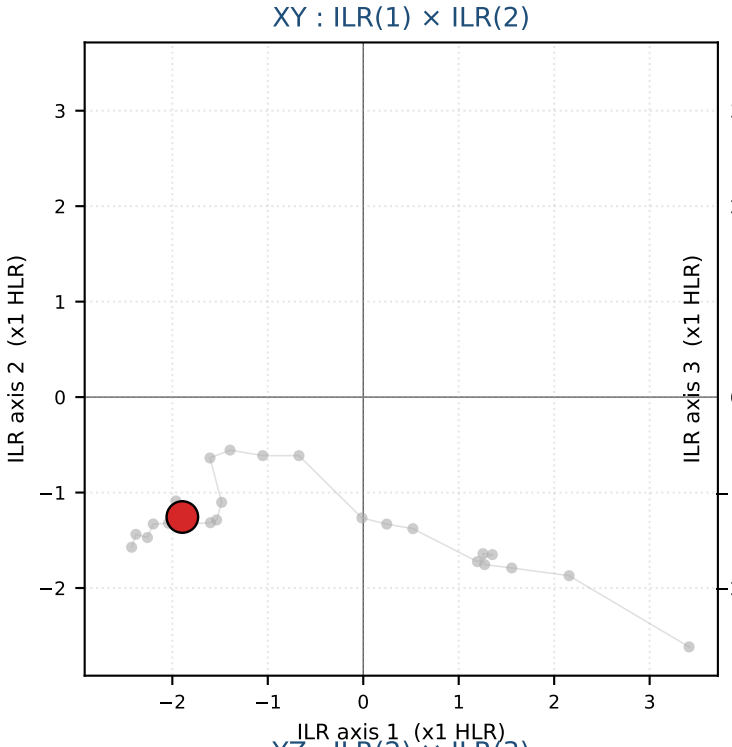
display factor: x1 (raw HLR x 1)

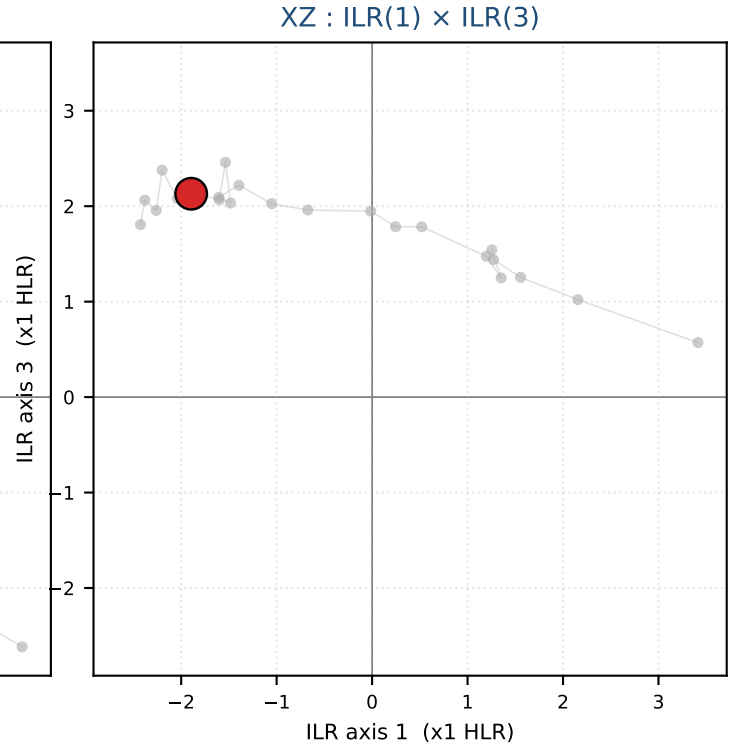
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2007 | D=9 T=26 | level 1 first-principles

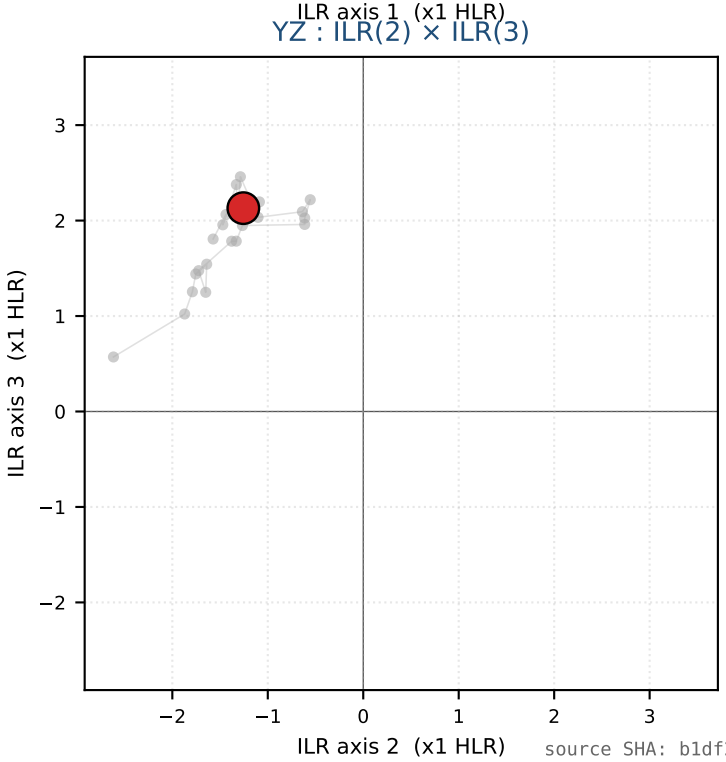
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

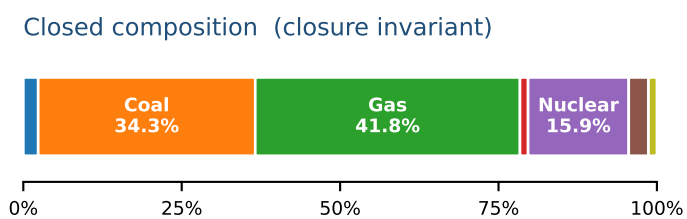




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0235	+4.115
Coal	0.3426	+6.795
Gas	0.4178	+6.993
Hydro	0.0128	+3.508
Nuclear	0.1588	+6.026
Other Fossil	0.0312	+4.398
Other Renewabl	0.0000	-32.656
Solar	0.0000	-2.723
Wind	0.0133	+3.545





closure: Σ = 1.000000 OK

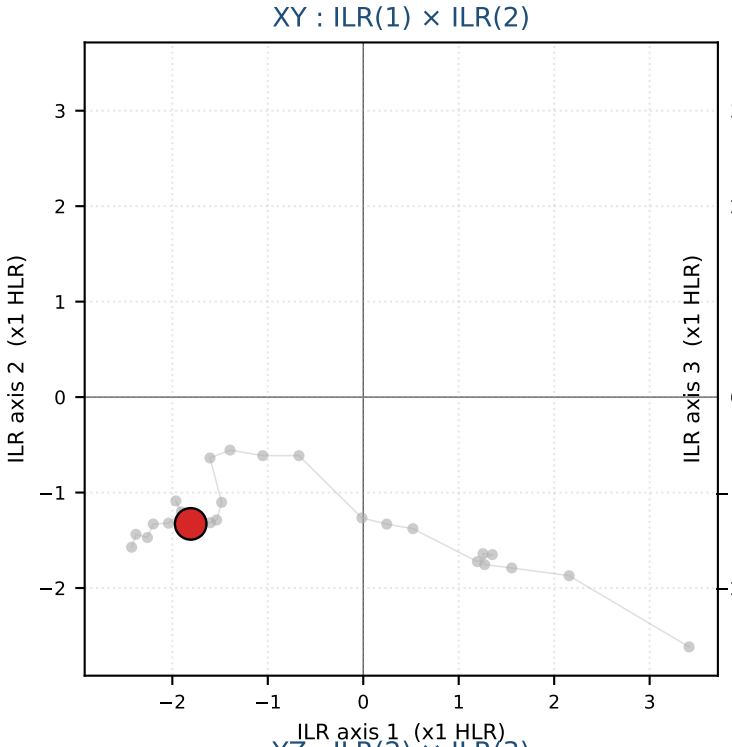
display factor: x1 (raw HLR x 1)

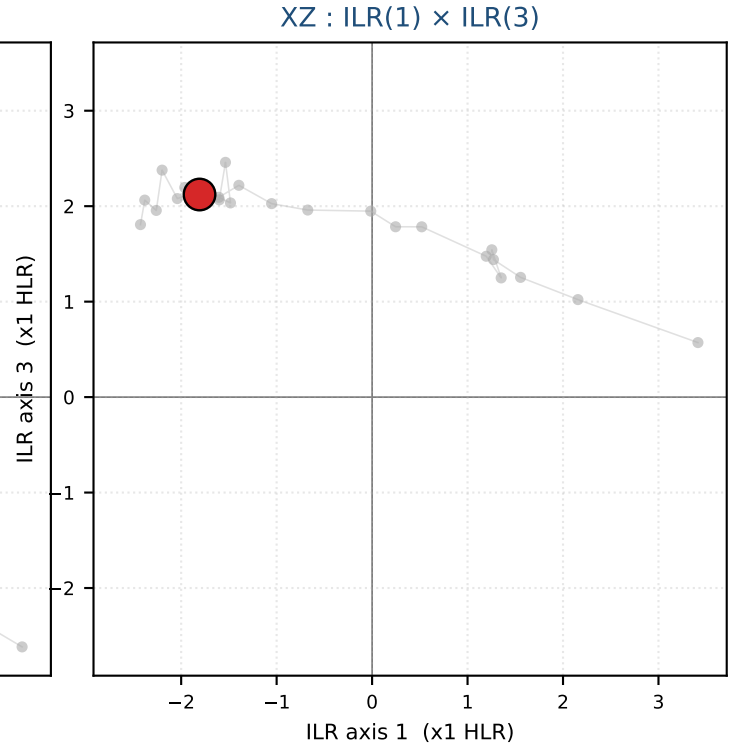
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2008 | D=9 T=26 | level 1 first-principles

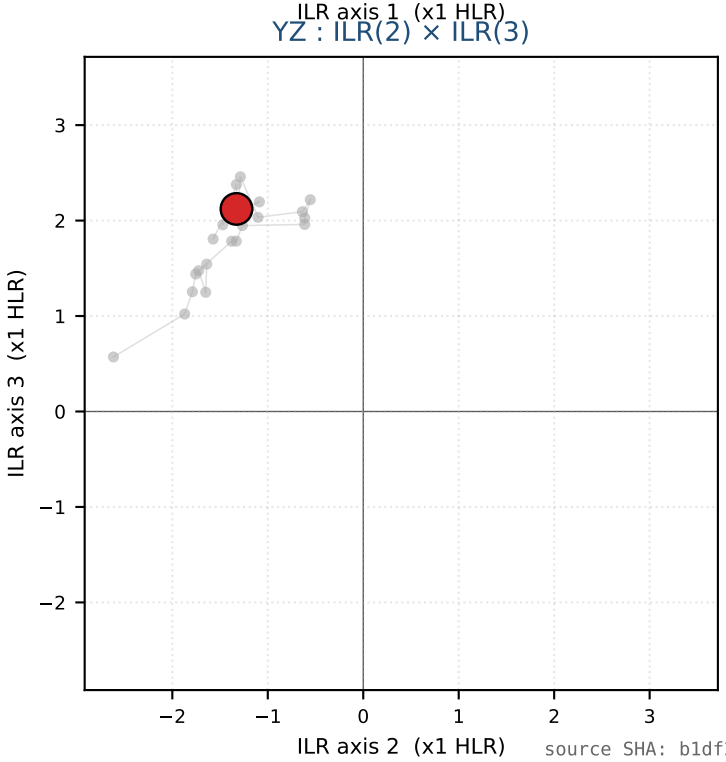
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

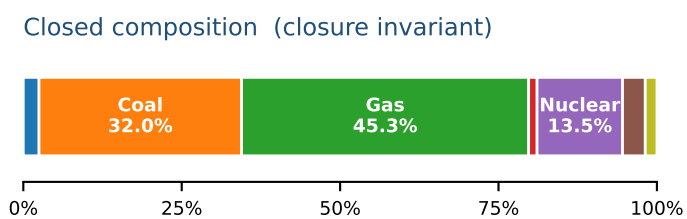




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0248	+4.044
Coal	0.3198	+6.600
Gas	0.4531	+6.949
Hydro	0.0132	+3.414
Nuclear	0.1349	+5.738
Other Fossil	0.0358	+4.411
Other Renewabl	0.0000	-32.762
Solar	0.0001	-2.135
Wind	0.0183	+3.741





closure: Σ = 1.000000 OK

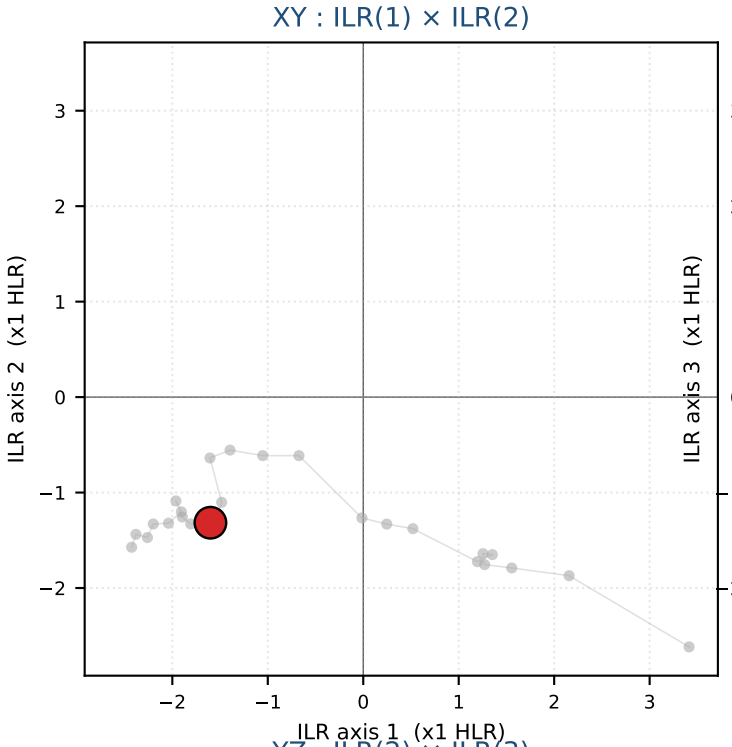
display factor: x1 (raw HLR x 1)

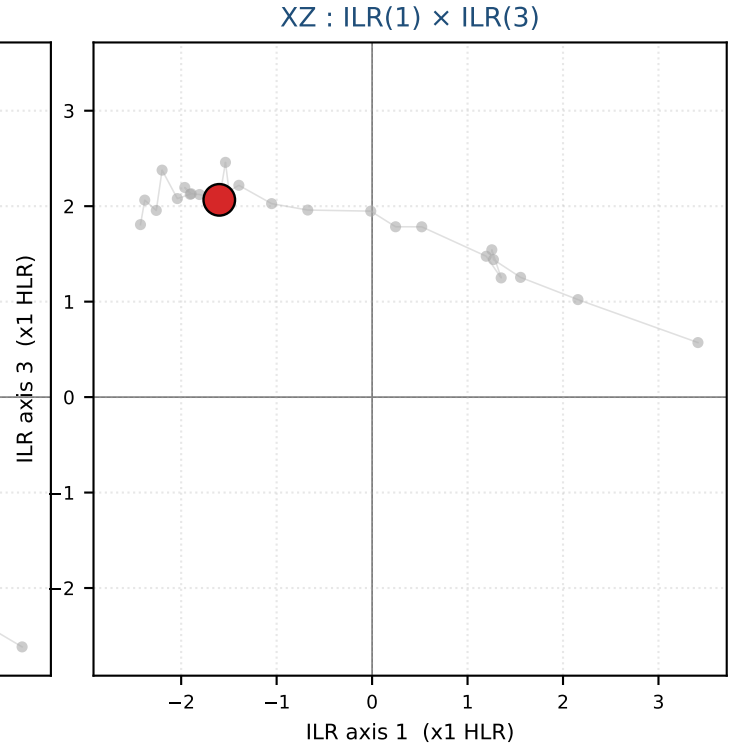
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2009 | D=9 T=26 | level 1 first-principles

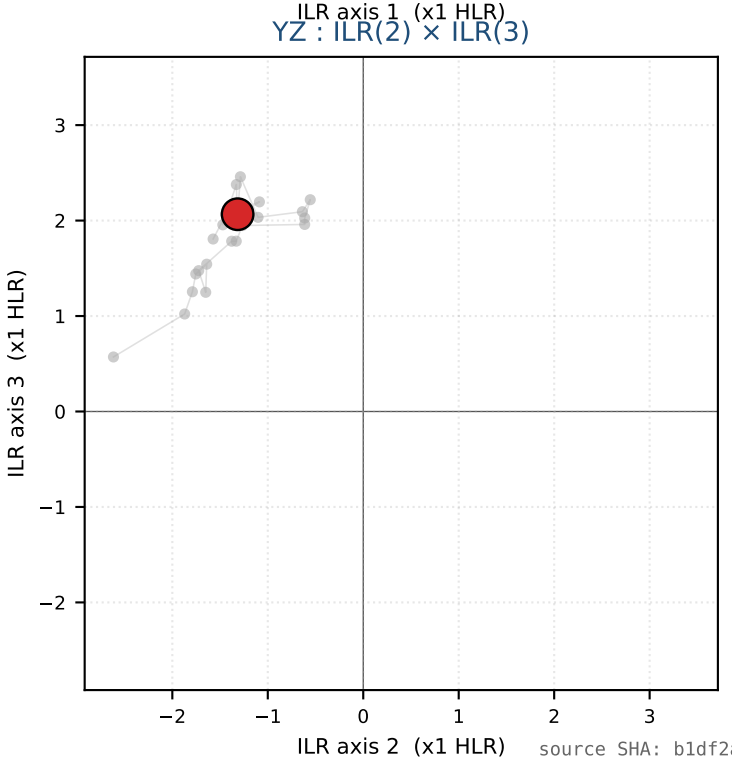
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

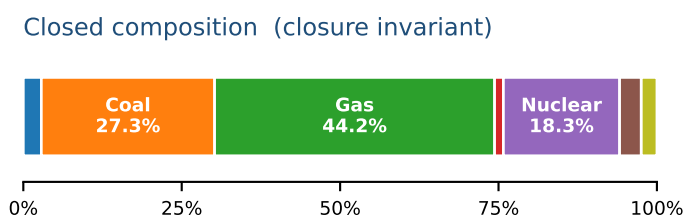




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0284	+4.111
Coal	0.2735	+6.375
Gas	0.4419	+6.855
Hydro	0.0139	+3.394
Nuclear	0.1834	+5.975
Other Fossil	0.0342	+4.295
Other Renewabl	0.0000	-32.799
Solar	0.0001	-2.172
Wind	0.0246	+3.967





closure: Σ = 1.000000 OK

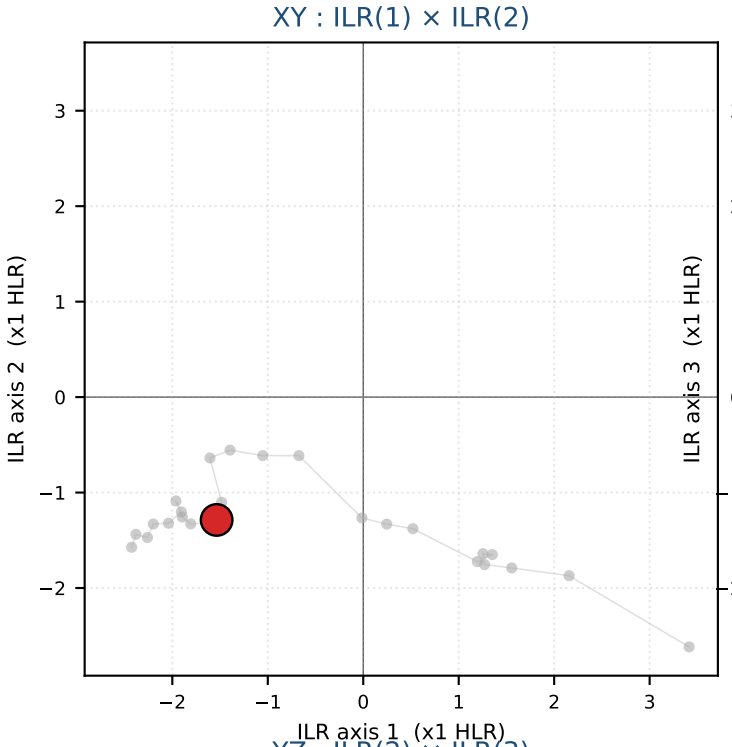
display factor: x1 (raw HLR x 1)

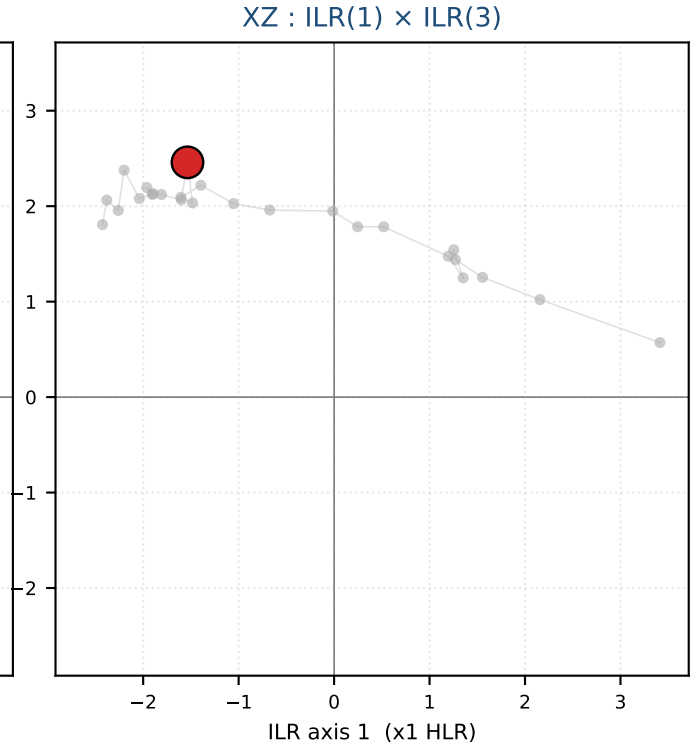
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2010 | D=9 T=26 | level 1 first-principles

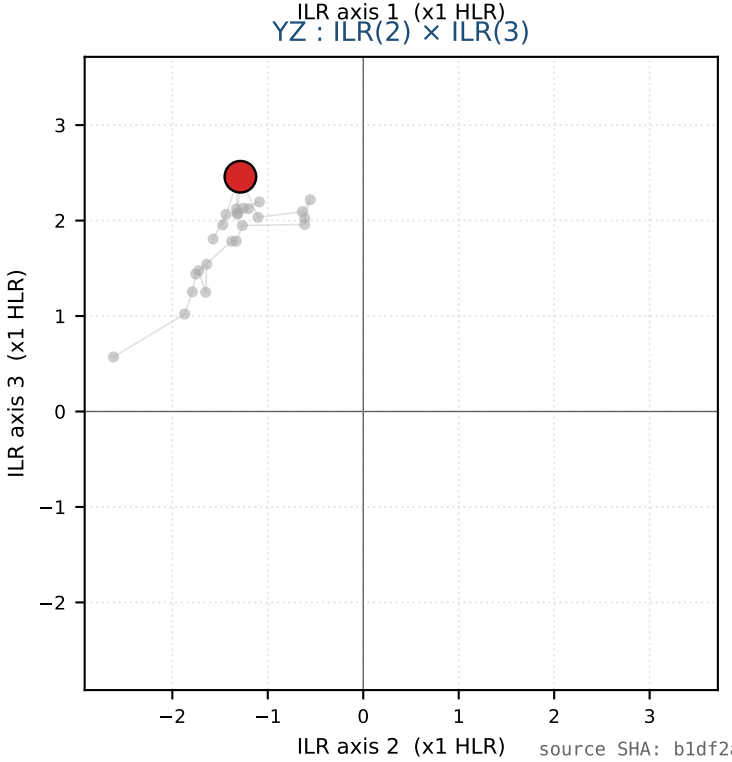
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



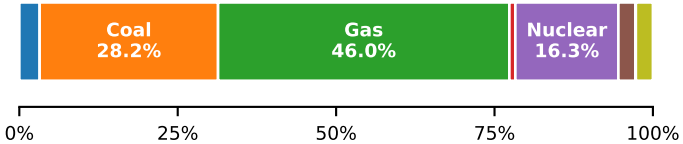


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0321	+4.208
Coal	0.2816	+6.380
Gas	0.4597	+6.870
Hydro	0.0094	+2.980
Nuclear	0.1626	+5.831
Other Fossil	0.0275	+4.053
Other Renewabl	0.0000	-32.837
Solar	0.0001	-1.517
Wind	0.0269	+4.033



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

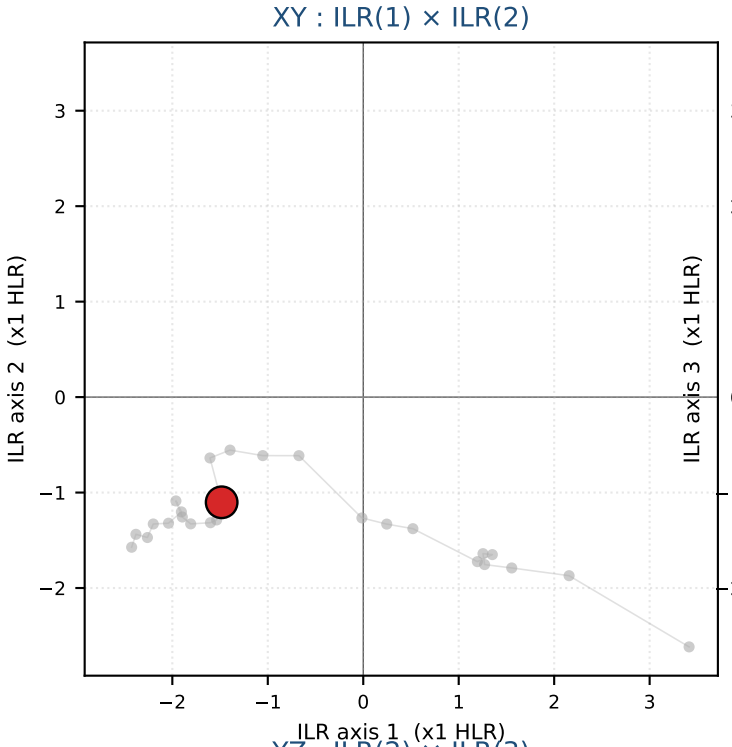
display factor: x1 (raw HLR x 1)

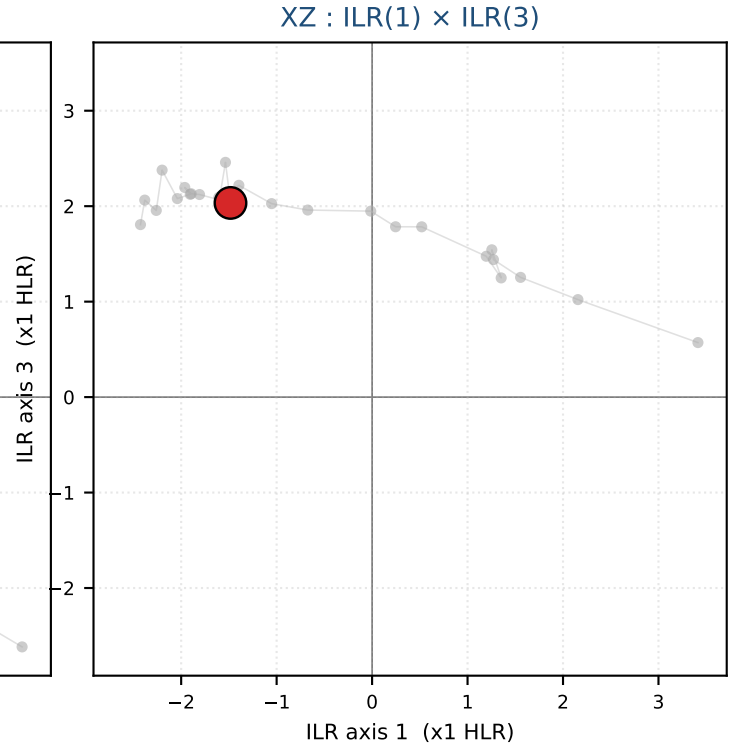
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2011 | D=9 T=26 | level 1 first-principles

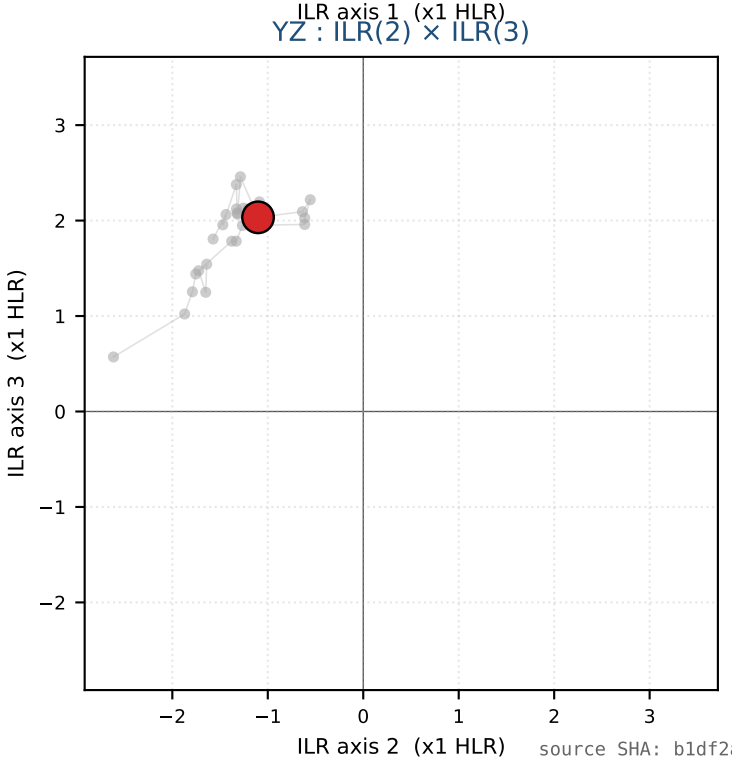
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

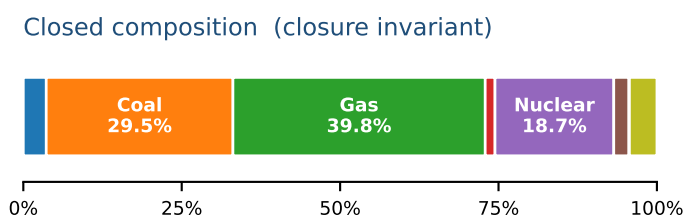




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0362	+4.009
Coal	0.2947	+6.106
Gas	0.3981	+6.407
Hydro	0.0155	+3.159
Nuclear	0.1875	+5.654
Other Fossil	0.0241	+3.601
Other Renewabl	0.0000	-33.119
Solar	0.0007	-0.007
Wind	0.0434	+4.190





closure: Σ = 1.000000 OK

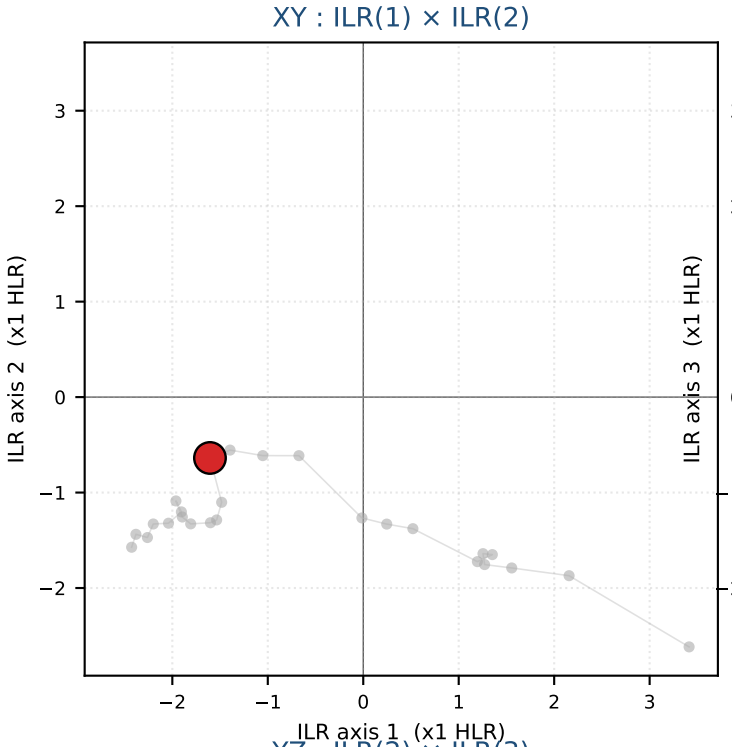
display factor: x1 (raw HLR x 1)

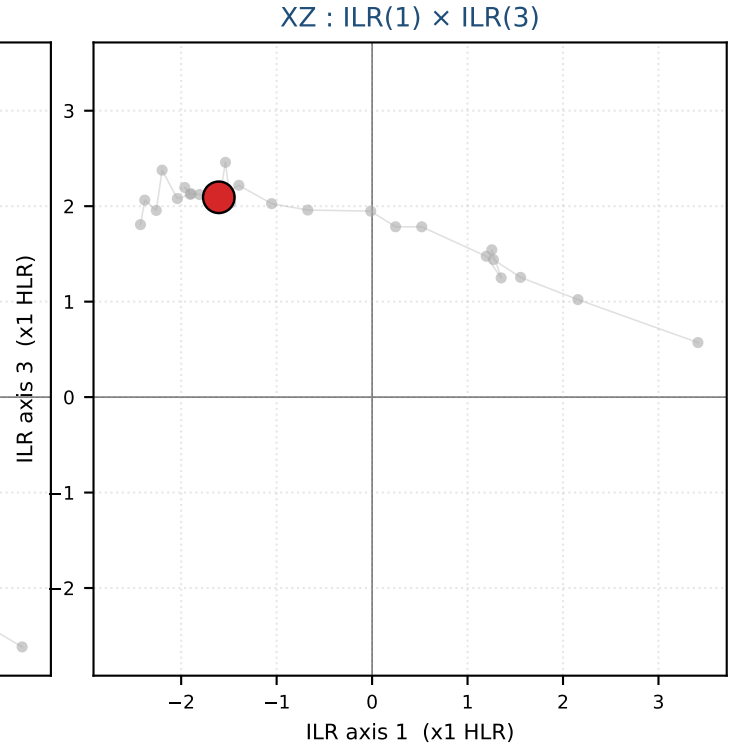
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2012 | D=9 T=26 | level 1 first-principles

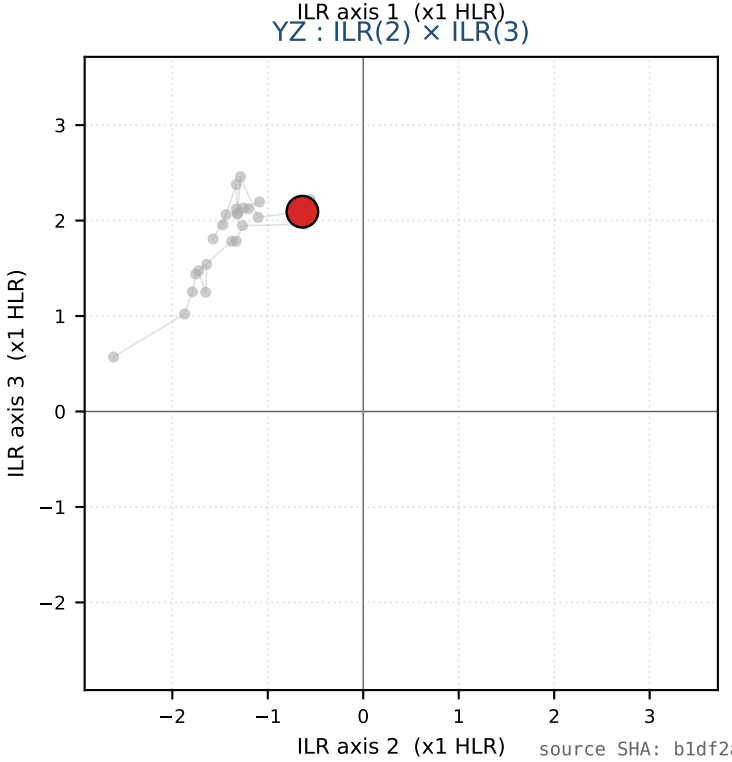
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

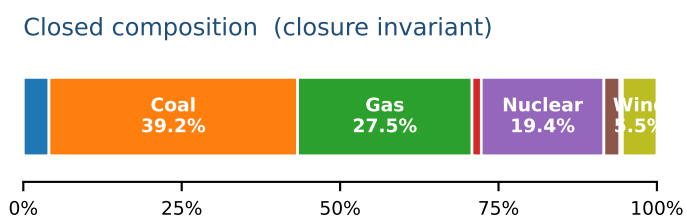




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0405	+3.895
Coal	0.3924	+6.166
Gas	0.2753	+5.812
Hydro	0.0146	+2.874
Nuclear	0.1935	+5.459
Other Fossil	0.0254	+3.430
Other Renewabl	0.0000	-33.334
Solar	0.0037	+1.505
Wind	0.0545	+4.192





closure: Σ = 1.000000 OK

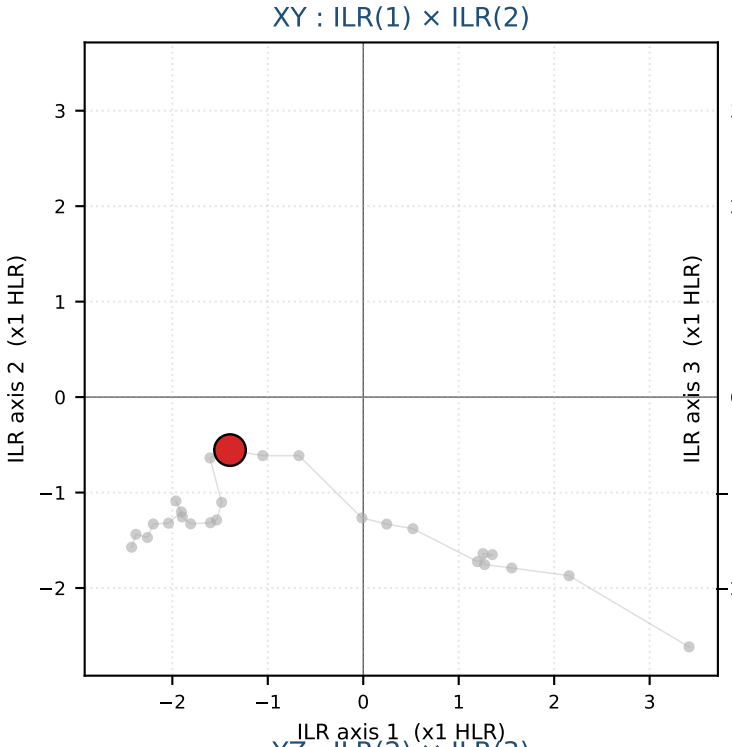
display factor: x1 (raw HLR x 1)

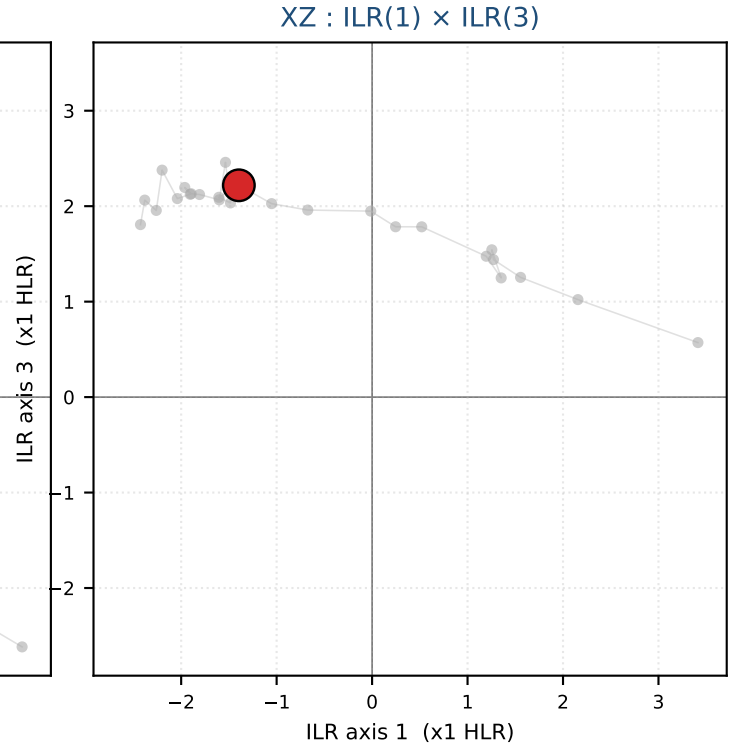
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2013 | D=9 T=26 | level 1 first-principles

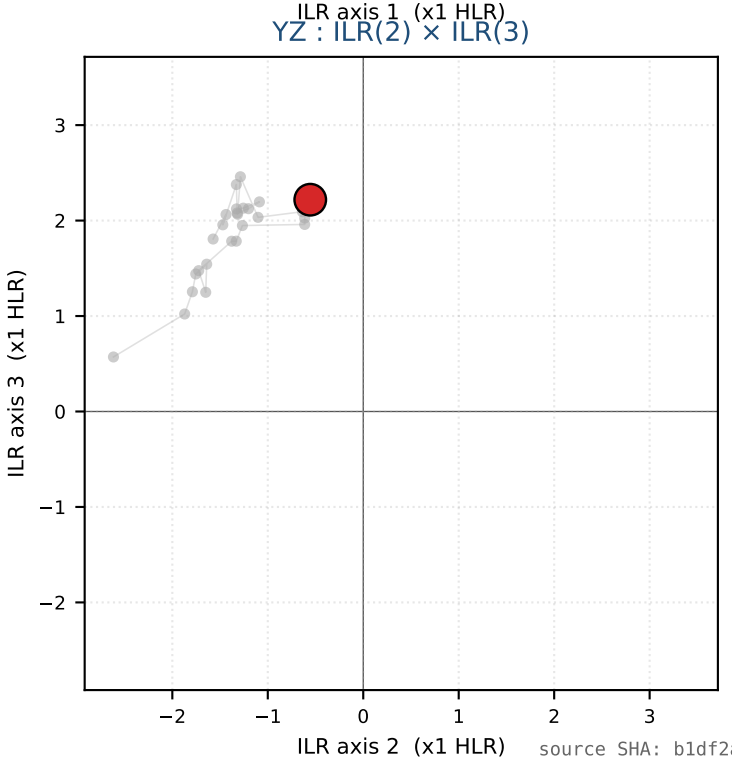
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

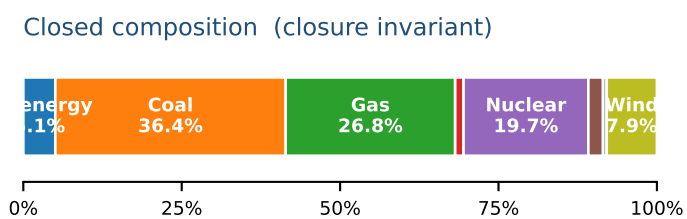




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0505	+4.033
Coal	0.3636	+6.007
Gas	0.2675	+5.700
Hydro	0.0131	+2.685
Nuclear	0.1971	+5.395
Other Fossil	0.0233	+3.261
Other Renewabl	0.0000	-33.401
Solar	0.0056	+1.836
Wind	0.0793	+4.484





closure: Σ = 1.000000 OK

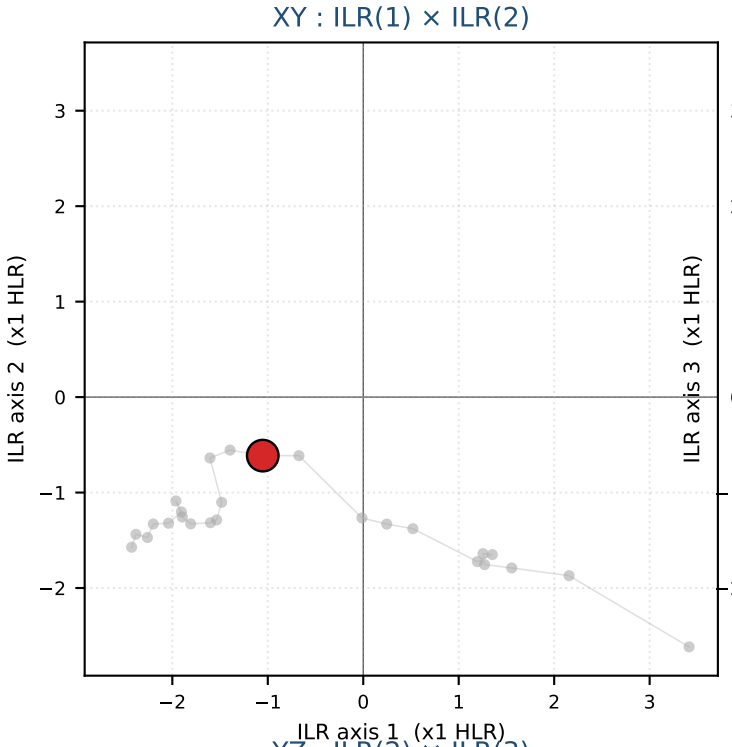
display factor: x1 (raw HLR x 1)

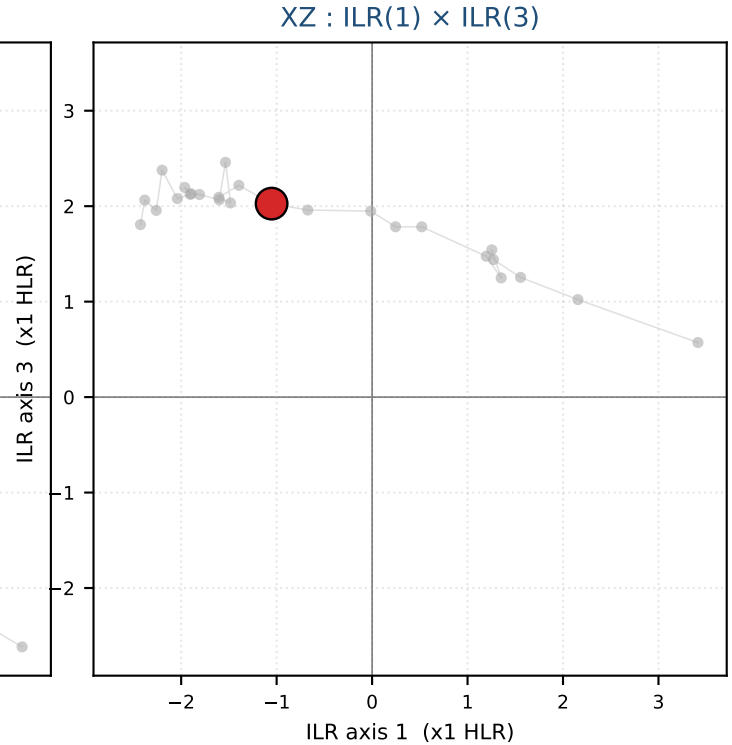
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2014 | D=9 T=26 | level 1 first-principles

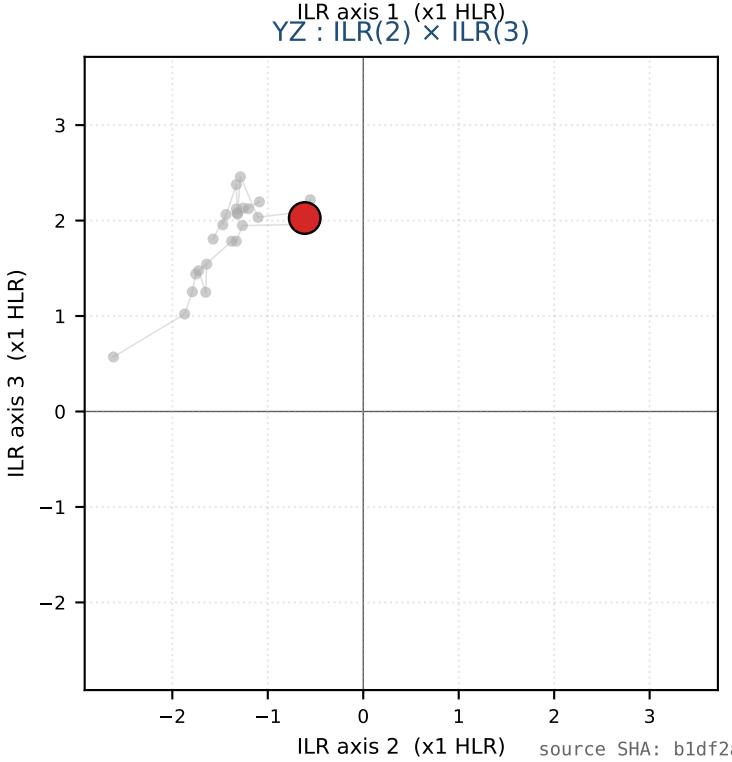
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



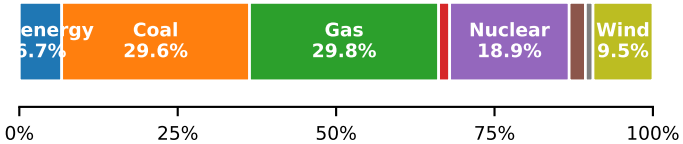


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0669	+4.146
Coal	0.2965	+5.635
Gas	0.2984	+5.641
Hydro	0.0174	+2.800
Nuclear	0.1886	+5.182
Other Fossil	0.0257	+3.189
Other Renewabl	0.0000	-33.512
Solar	0.0120	+2.426
Wind	0.0945	+4.491



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

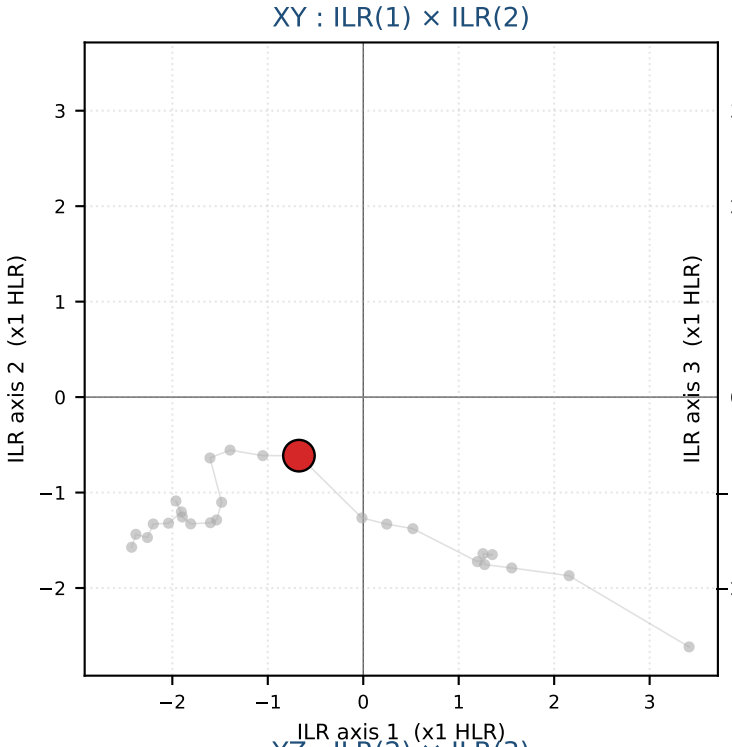
display factor: x1 (raw HLR x 1)

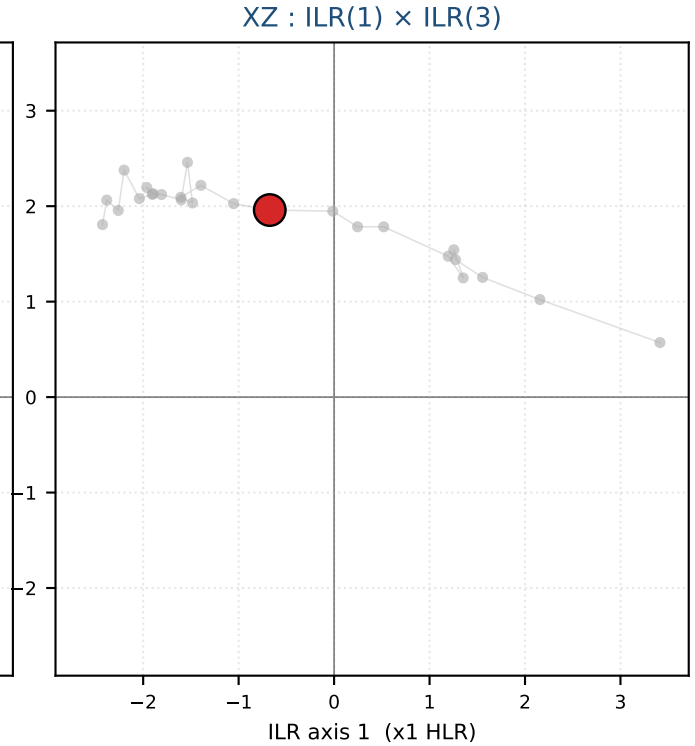
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2015 | D=9 T=26 | level 1 first-principles

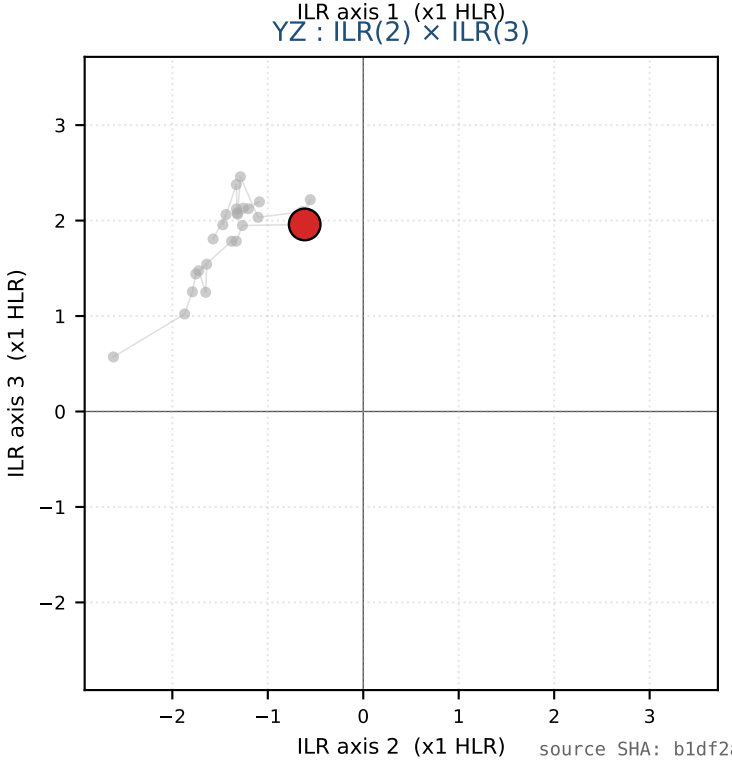
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

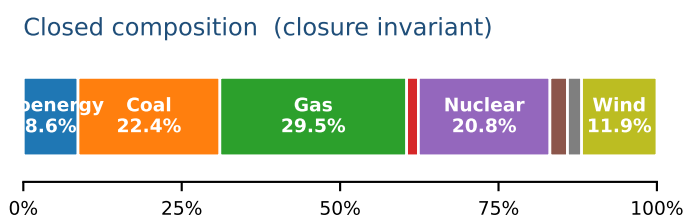




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0863	+4.285
Coal	0.2239	+5.238
Gas	0.2947	+5.513
Hydro	0.0186	+2.749
Nuclear	0.2076	+5.162
Other Fossil	0.0278	+3.151
Other Renewabl	0.0000	-33.630
Solar	0.0222	+2.928
Wind	0.1188	+4.604





closure: Σ = 1.000000 OK

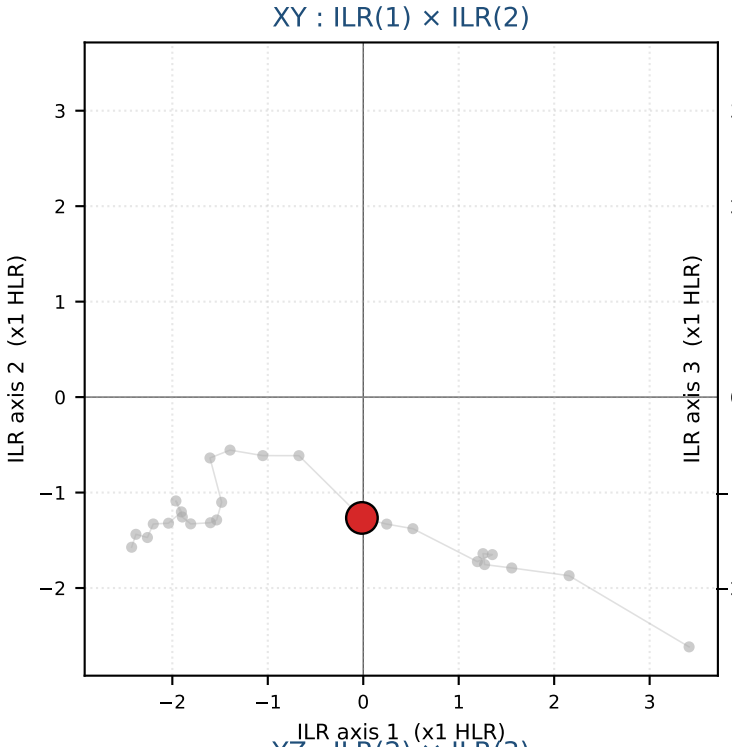
display factor: x1 (raw HLR x 1)

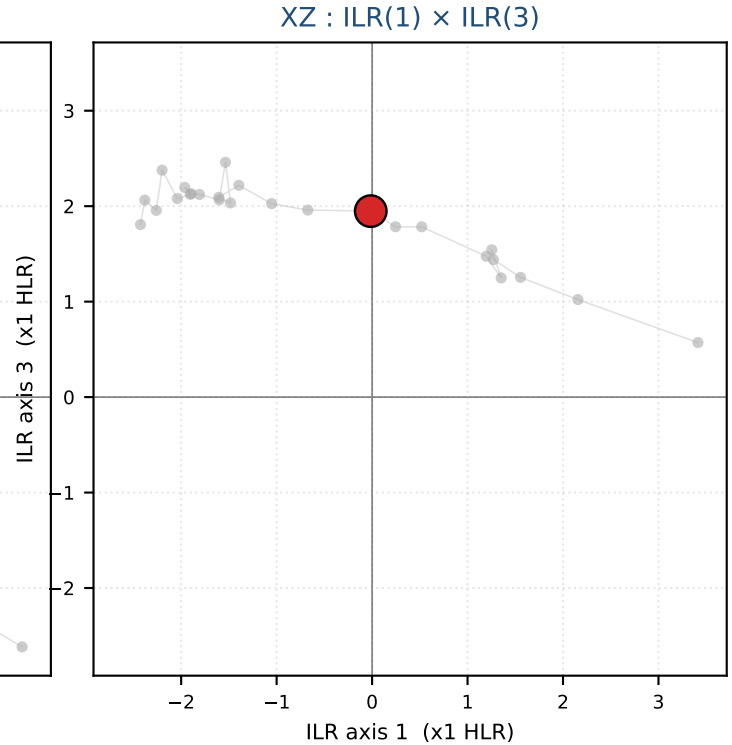
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2016 | D=9 T=26 | level 1 first-principles

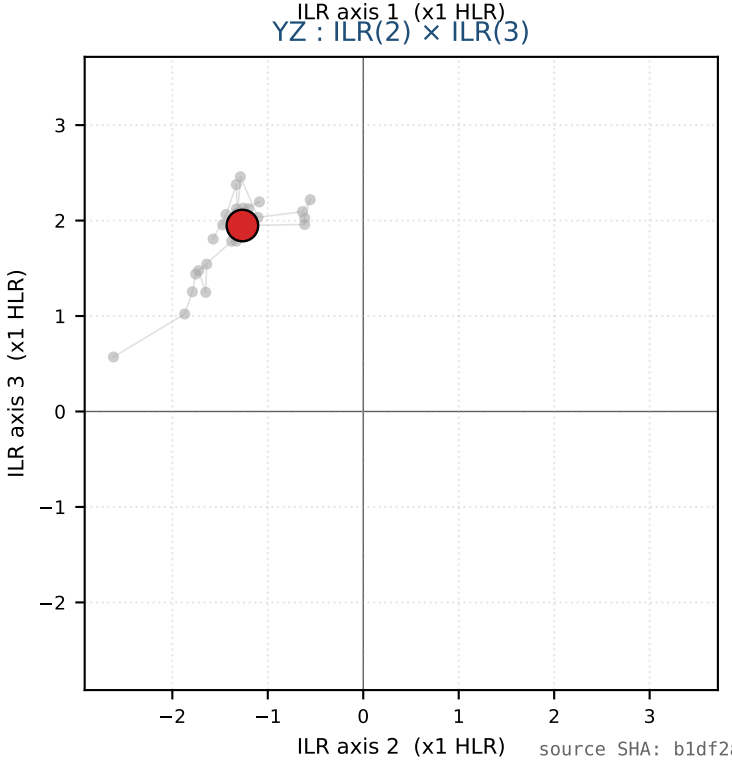
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

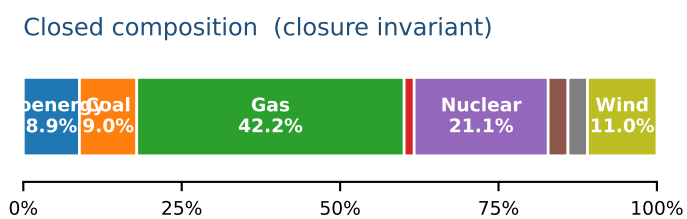




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0887	+4.345
Coal	0.0904	+4.365
Gas	0.4220	+5.905
Hydro	0.0158	+2.622
Nuclear	0.2115	+5.214
Other Fossil	0.0314	+3.306
Other Renewabl	0.0000	-33.597
Solar	0.0307	+3.284
Wind	0.1096	+4.557





closure: Σ = 1.000000 OK

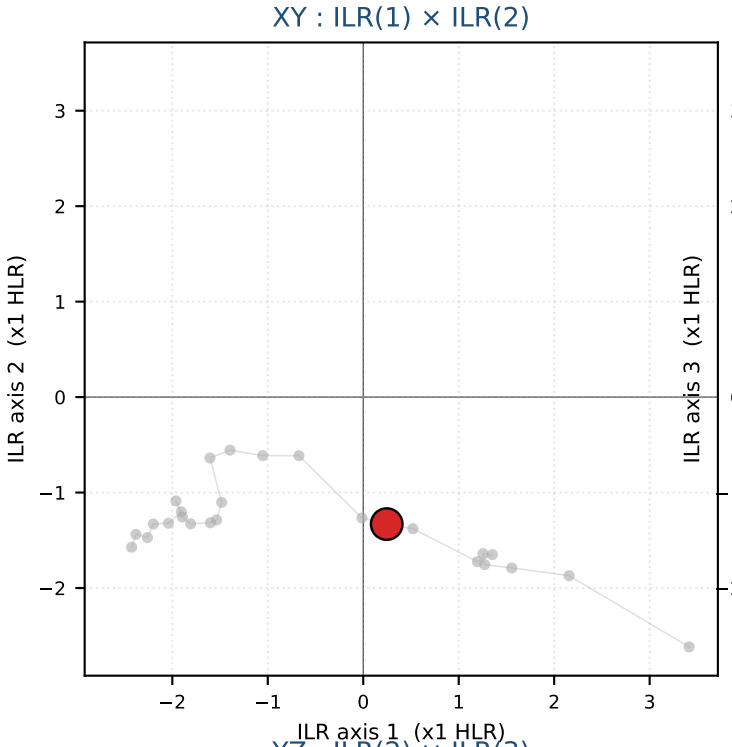
display factor: x1 (raw HLR x 1)

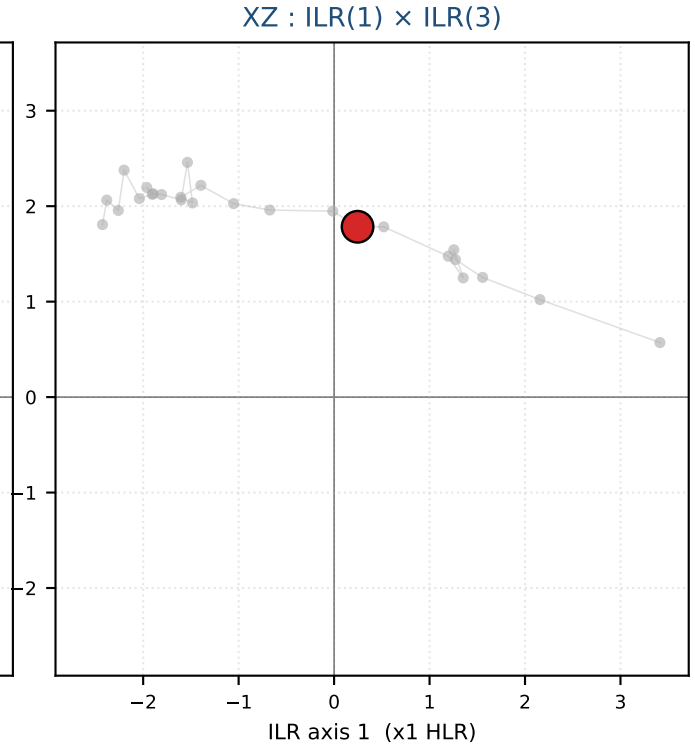
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2017 | D=9 T=26 | level 1 first-principles

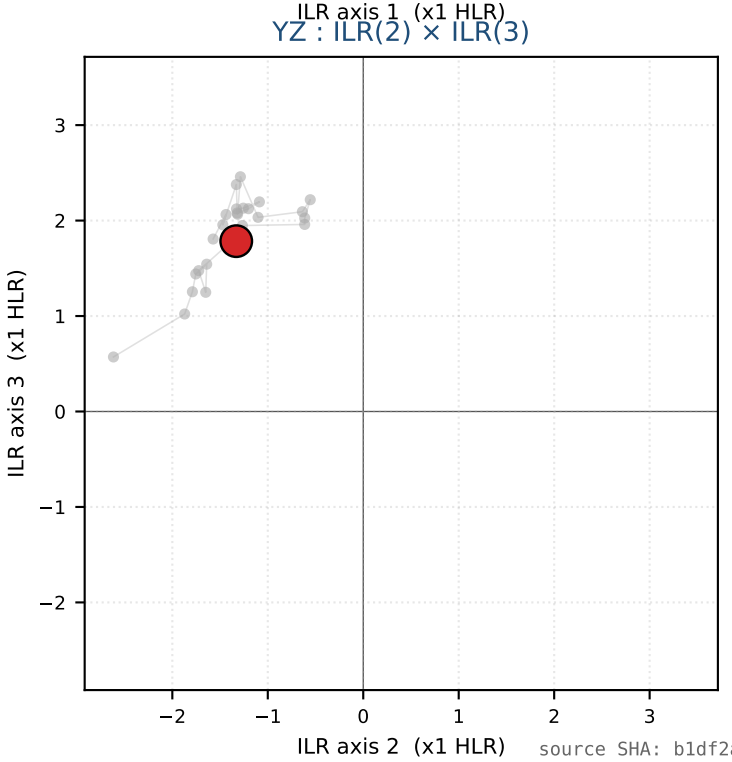
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

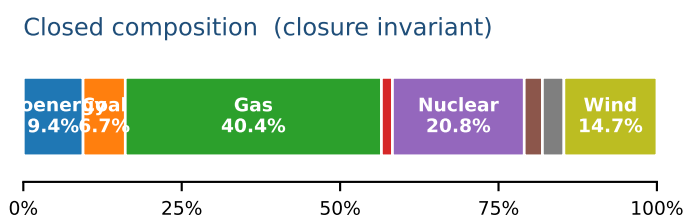




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0943	+4.396
Coal	0.0666	+4.048
Gas	0.4043	+5.852
Hydro	0.0174	+2.705
Nuclear	0.2080	+5.187
Other Fossil	0.0287	+3.207
Other Renewabl	0.0000	-33.605
Solar	0.0339	+3.372
Wind	0.1468	+4.838





closure: Σ = 1.000000 OK

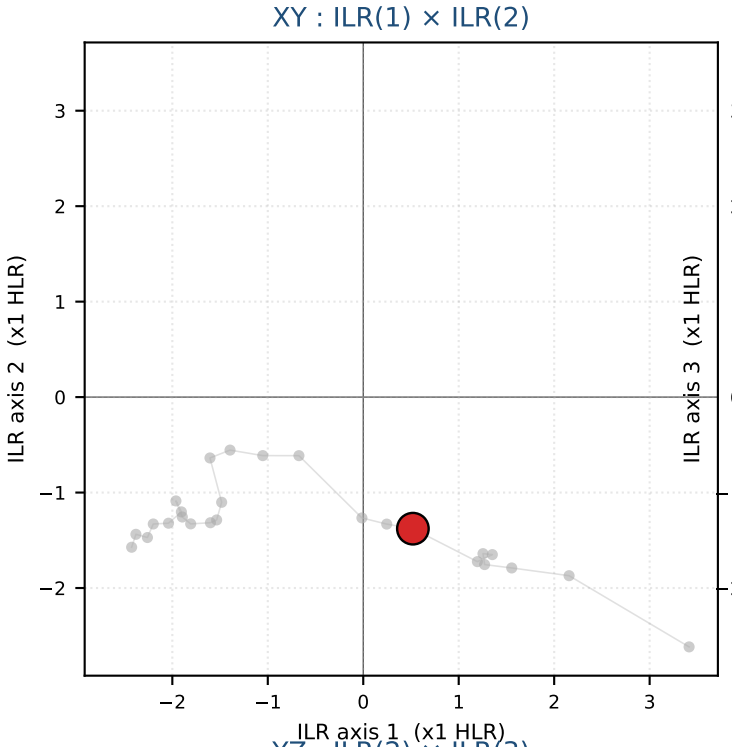
display factor: x1 (raw HLR x 1)

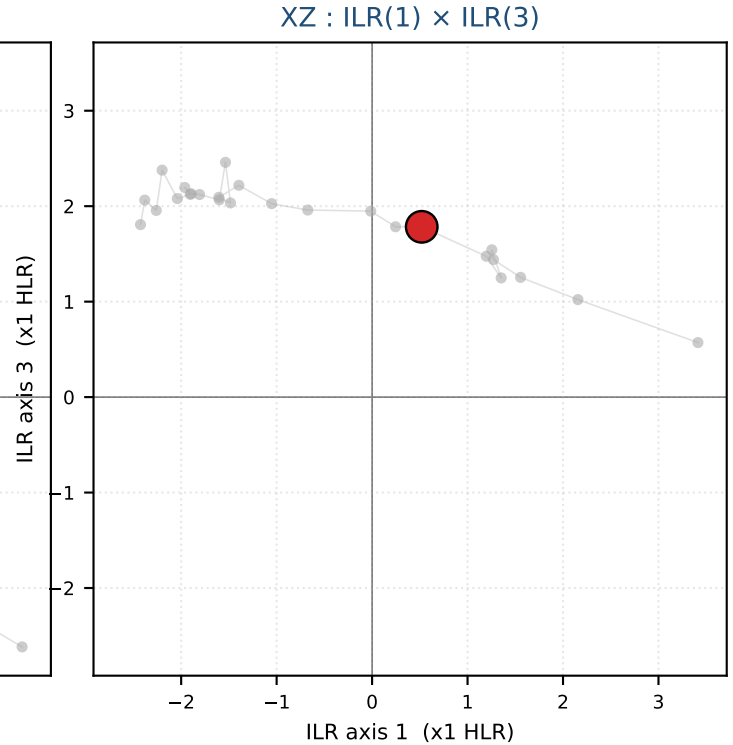
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2018 | D=9 T=26 | level 1 first-principles

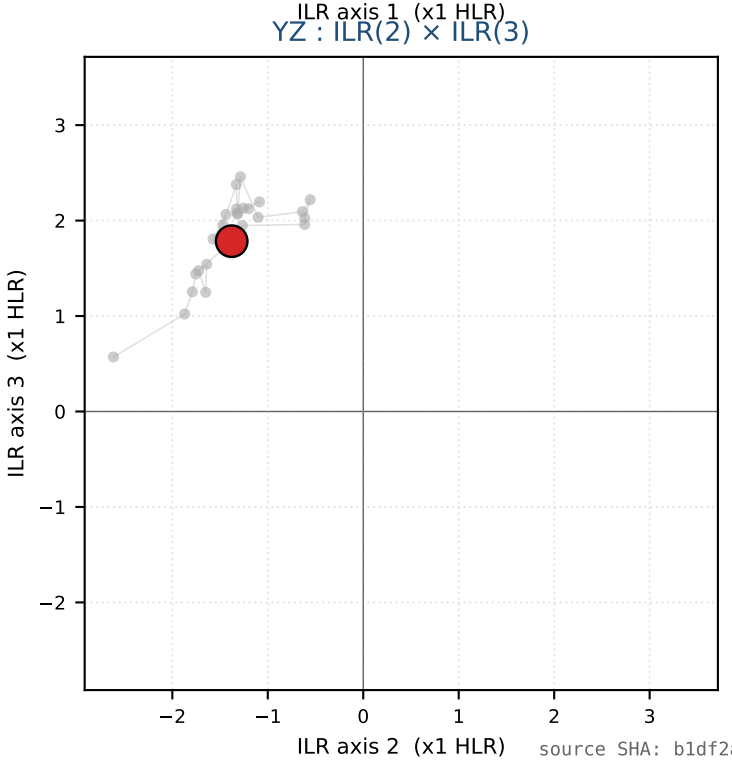
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



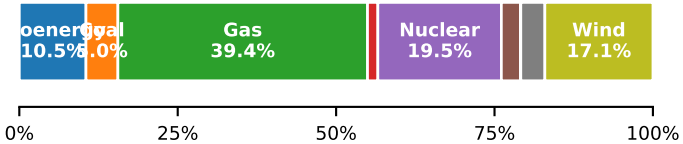


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1052	+1.177
Coal	0.0504	+0.442
Gas	0.3940	+2.498
Hydro	0.0163	-0.687
Nuclear	0.1949	+1.794
Other Fossil	0.0307	-0.055
Other Renewabl	0.0000	-6.986
Solar	0.0380	+0.158
Wind	0.1705	+1.660



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

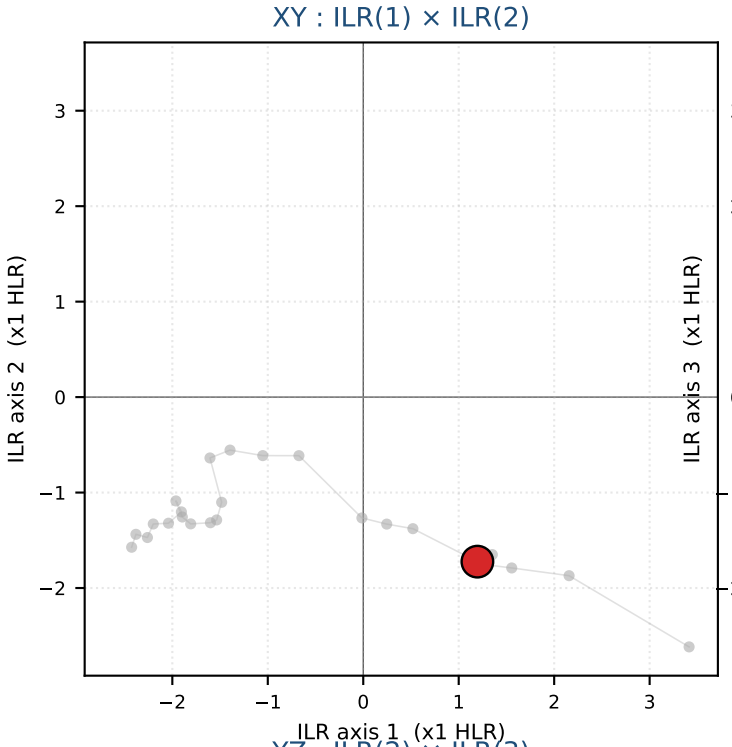
display factor: x1 (raw HLR x 1)

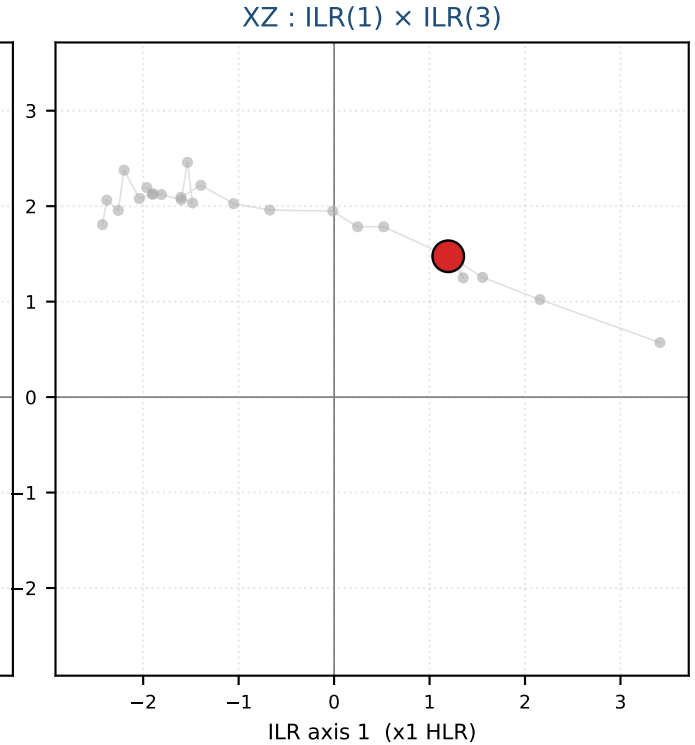
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2019 | D=9 T=26 | level 1 first-principles

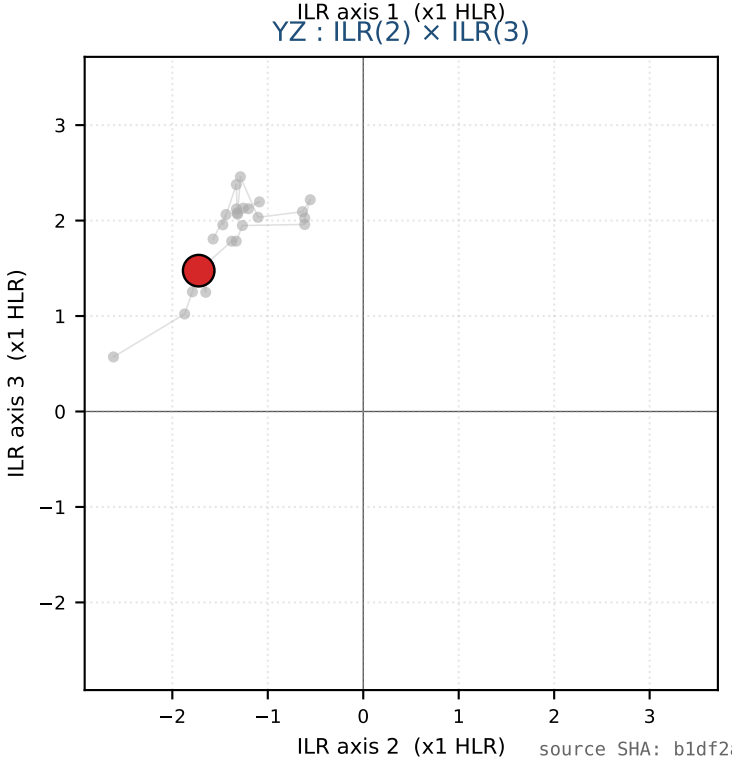
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

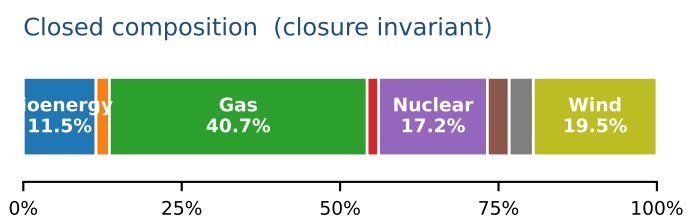




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1147	+1.320
Coal	0.0211	-0.371
Gas	0.4068	+2.585
Hydro	0.0181	-0.526
Nuclear	0.1717	+1.723
Other Fossil	0.0344	+0.116
Other Renewabl	0.0000	-6.911
Solar	0.0380	+0.214
Wind	0.1951	+1.851





closure: Σ = 1.000000 OK

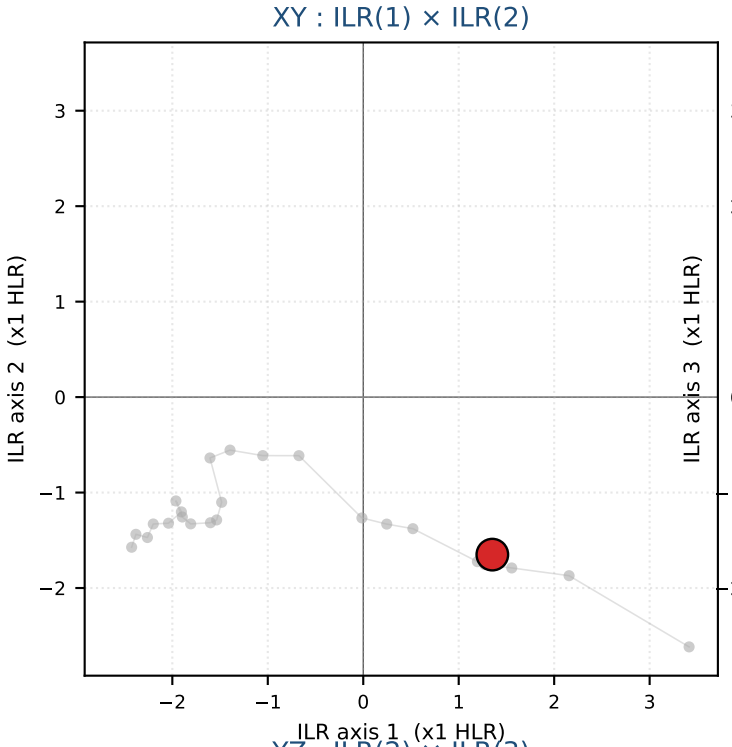
display factor: x1 (raw HLR x 1)

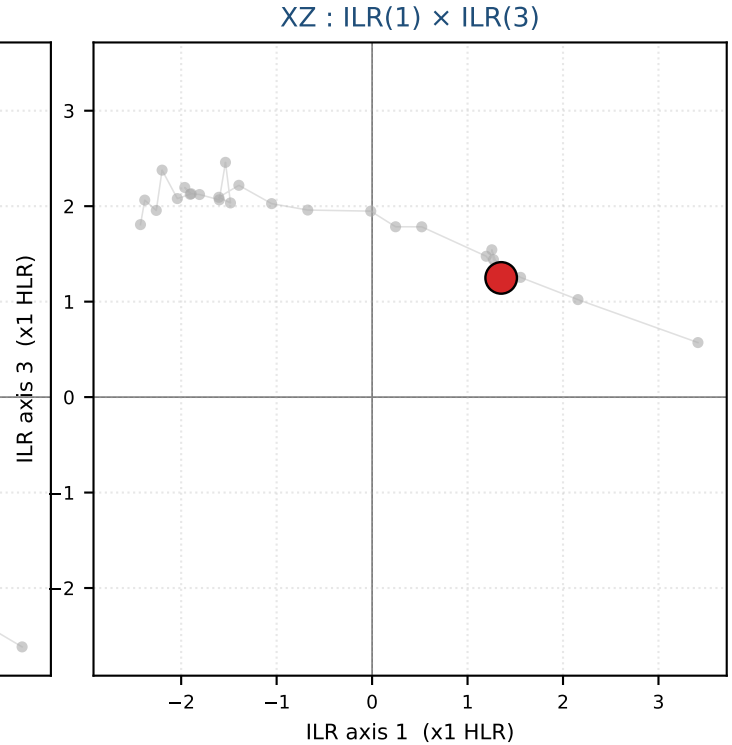
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2020 | D=9 T=26 | level 1 first-principles

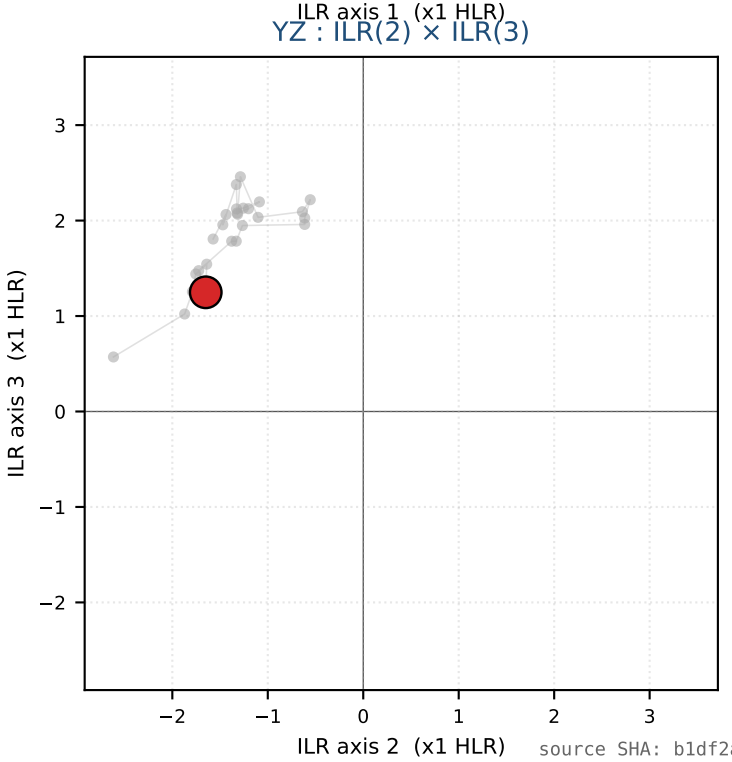
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

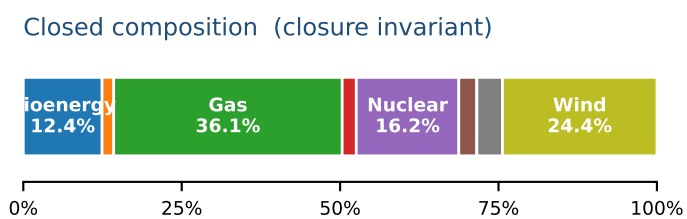




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1243	+4.714
Coal	0.0184	+2.801
Gas	0.3606	+5.778
Hydro	0.0222	+2.989
Nuclear	0.1619	+4.978
Other Fossil	0.0285	+3.240
Other Renewabl	0.0000	-33.478
Solar	0.0404	+3.591
Wind	0.2437	+5.387





closure: Σ = 1.000000 OK

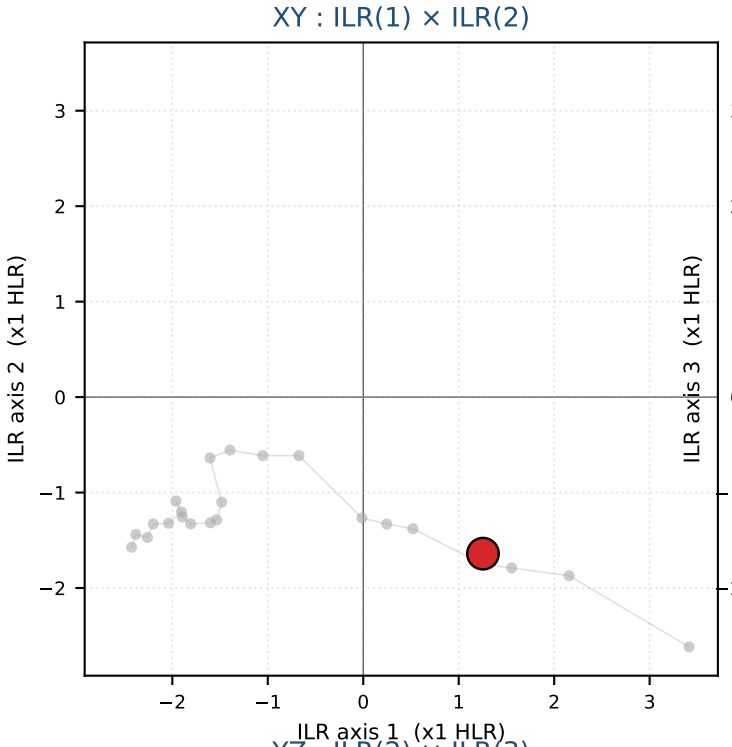
display factor: x1 (raw HLR x 1)

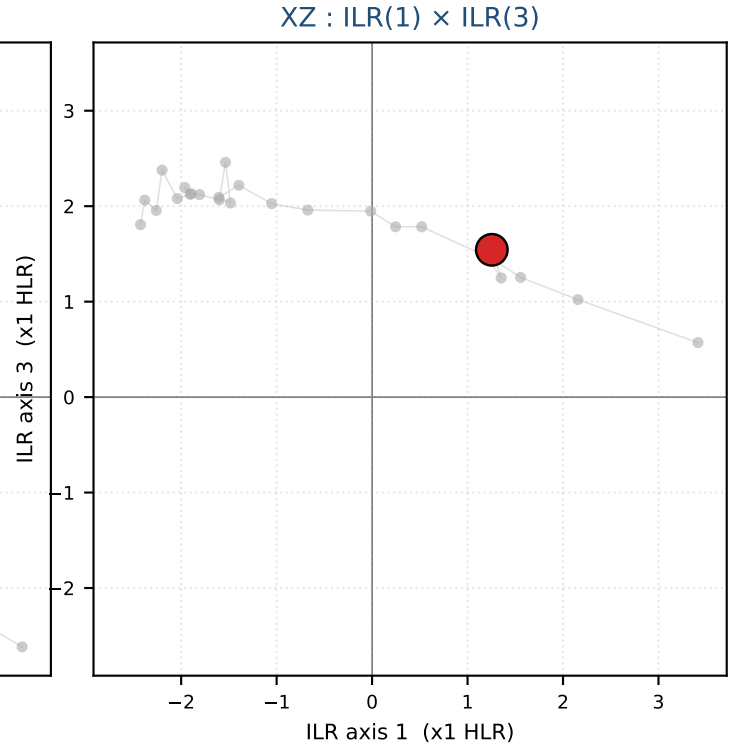
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2021 | D=9 T=26 | level 1 first-principles

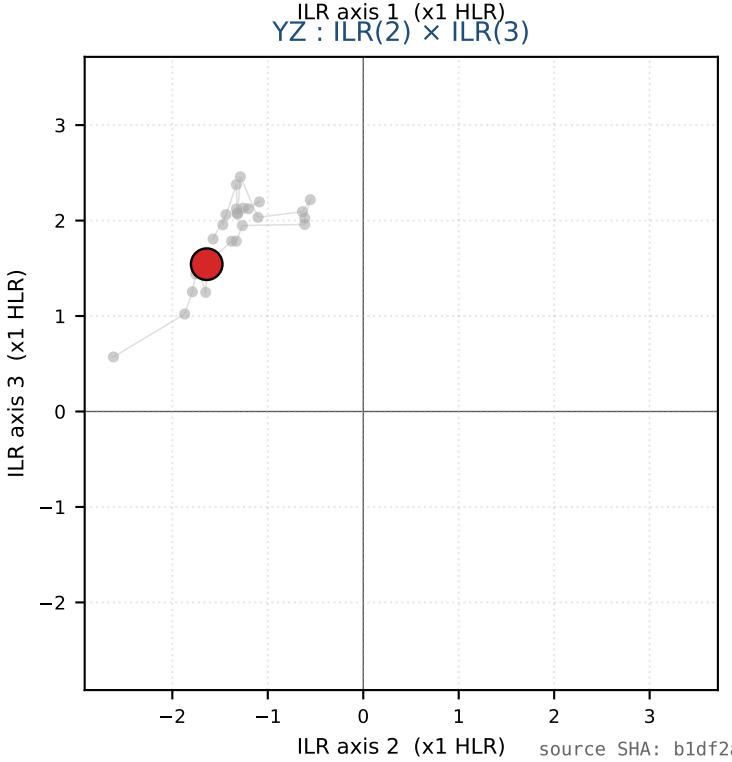
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

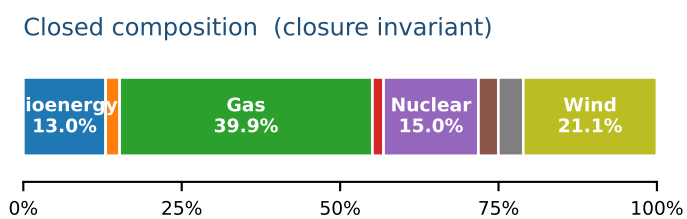




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1299	+4.763
Coal	0.0221	+2.989
Gas	0.3990	+5.885
Hydro	0.0176	+2.764
Nuclear	0.1497	+4.905
Other Fossil	0.0314	+3.344
Other Renewabl	0.0000	-33.465
Solar	0.0394	+3.569
Wind	0.2109	+5.247





closure: Σ = 1.000000 OK

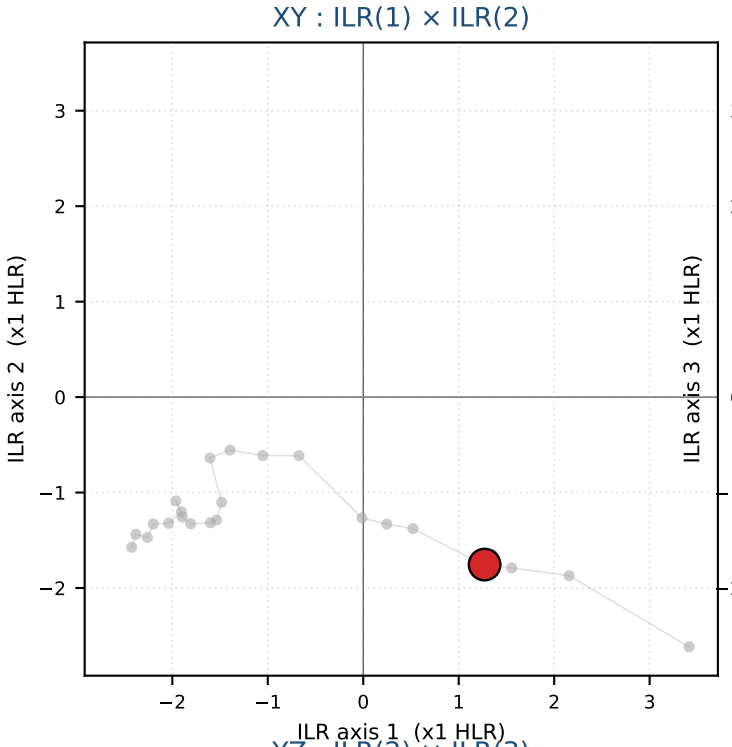
display factor: x1 (raw HLR x 1)

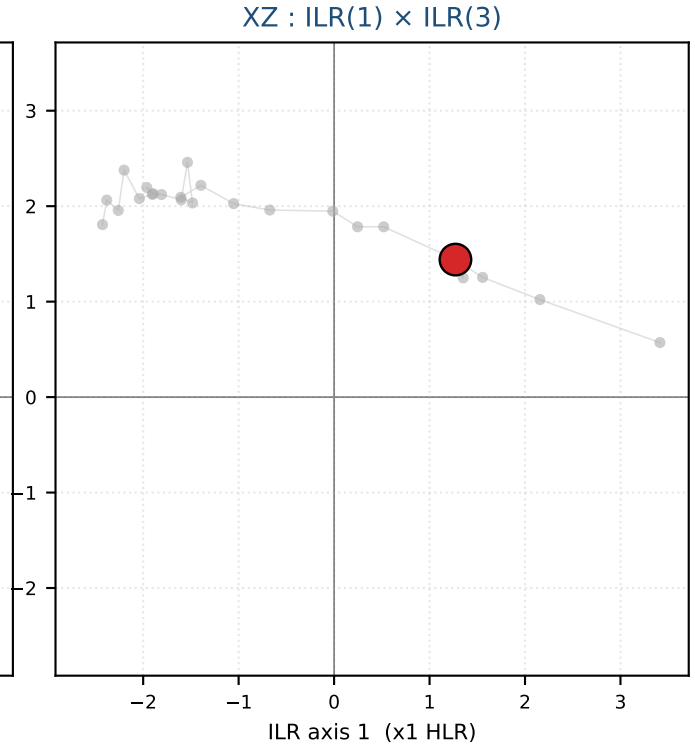
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2022 | D=9 T=26 | level 1 first-principles

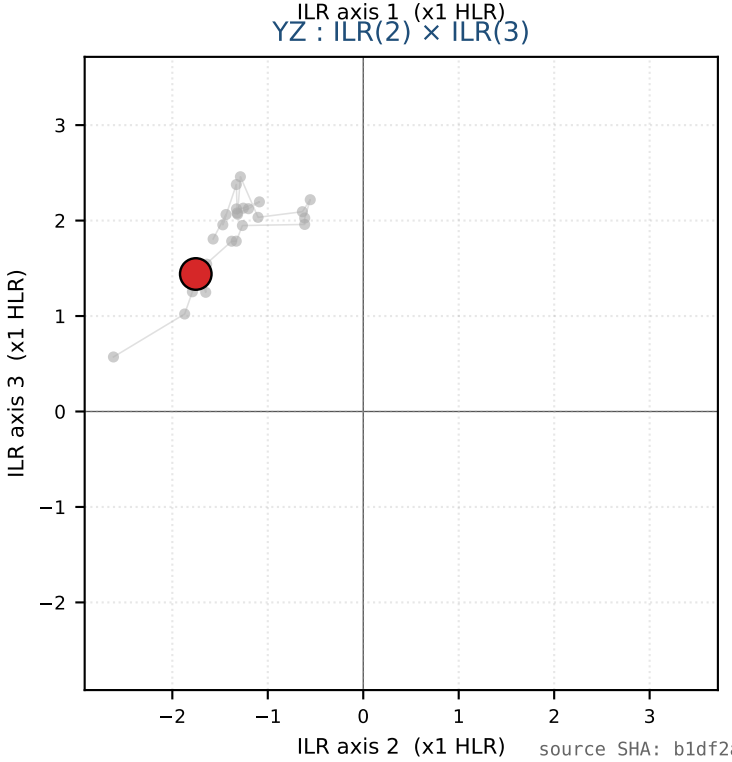
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

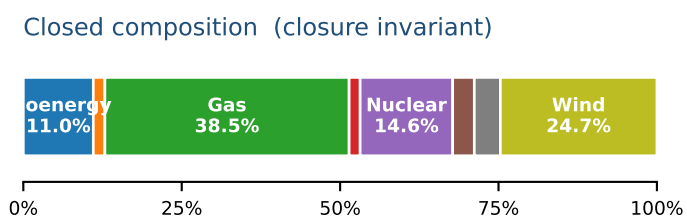




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1105	+4.621
Coal	0.0183	+2.824
Gas	0.3855	+5.871
Hydro	0.0174	+2.775
Nuclear	0.1460	+4.900
Other Fossil	0.0341	+3.446
Other Renewabl	0.0000	-33.497
Solar	0.0411	+3.633
Wind	0.2471	+5.427





closure: Σ = 1.000000 OK

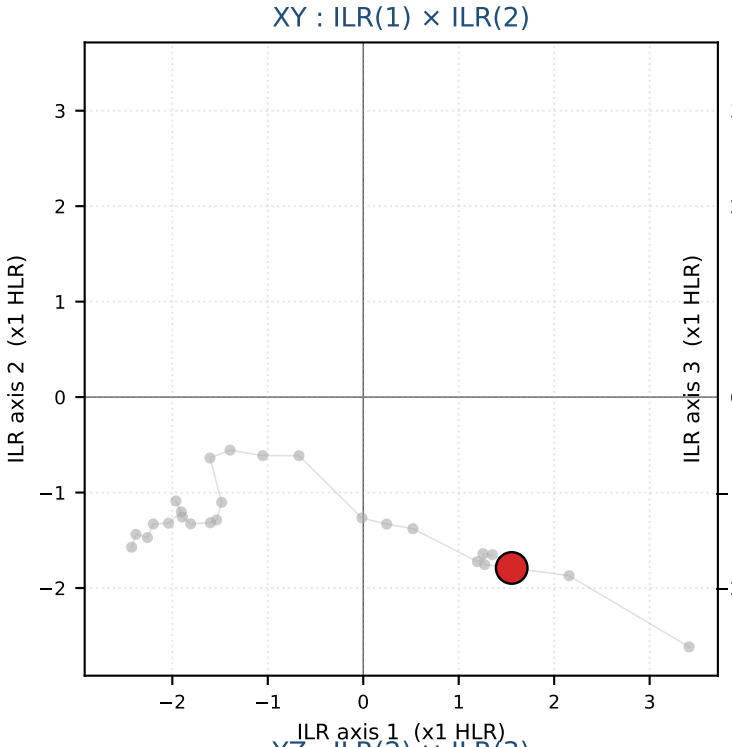
display factor: x1 (raw HLR x 1)

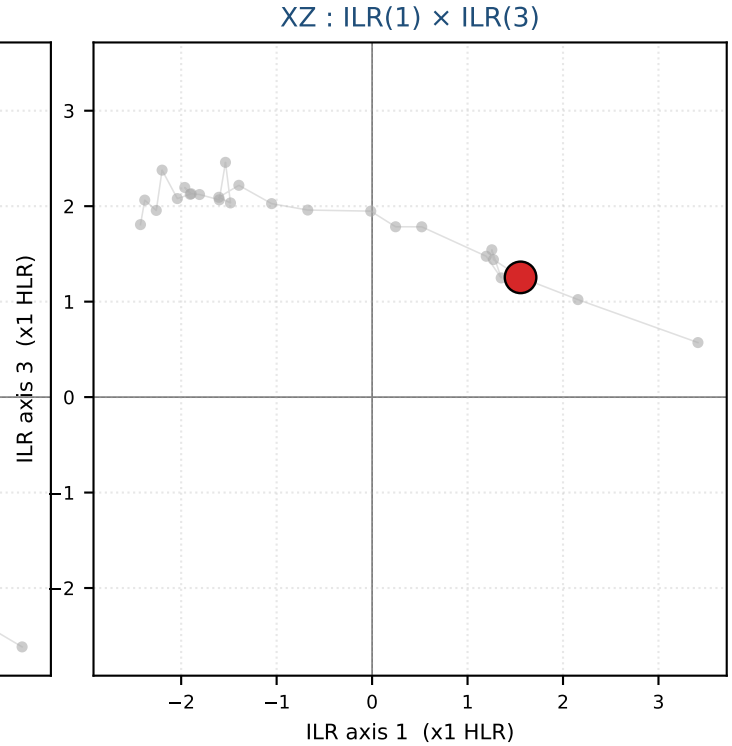
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2023 | D=9 T=26 | level 1 first-principles

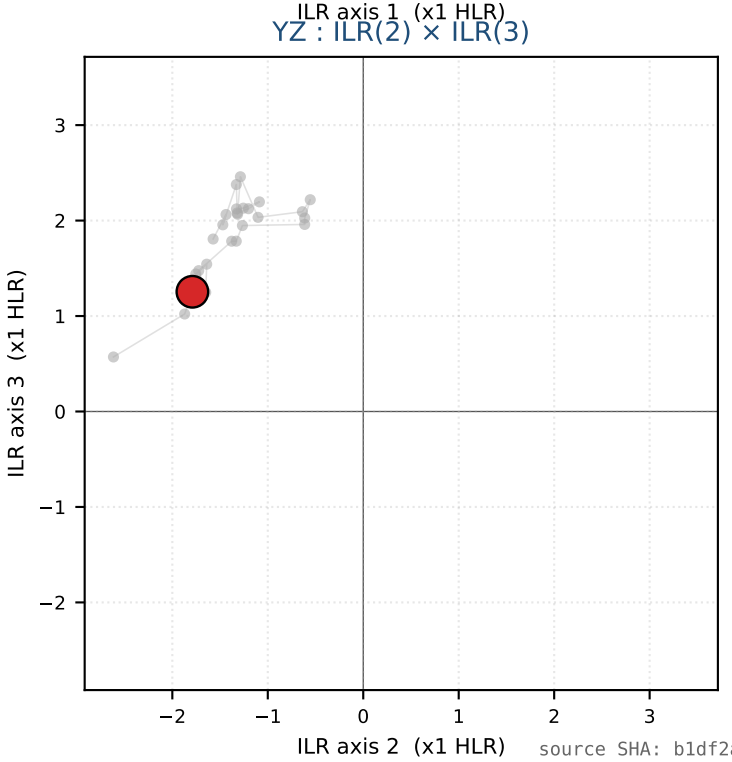
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

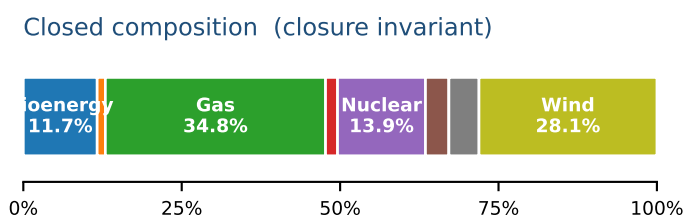




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1166	+4.665
Coal	0.0129	+2.465
Gas	0.3475	+5.757
Hydro	0.0189	+2.848
Nuclear	0.1388	+4.840
Other Fossil	0.0370	+3.517
Other Renewabl	0.0000	-33.403
Solar	0.0474	+3.766
Wind	0.2808	+5.544





closure: Σ = 1.000000 OK

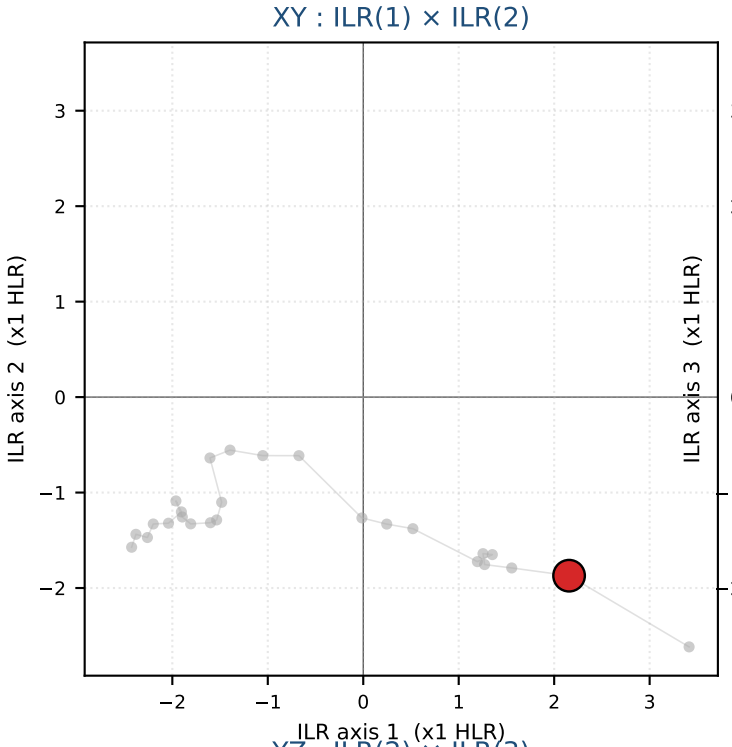
display factor: x1 (raw HLR x 1)

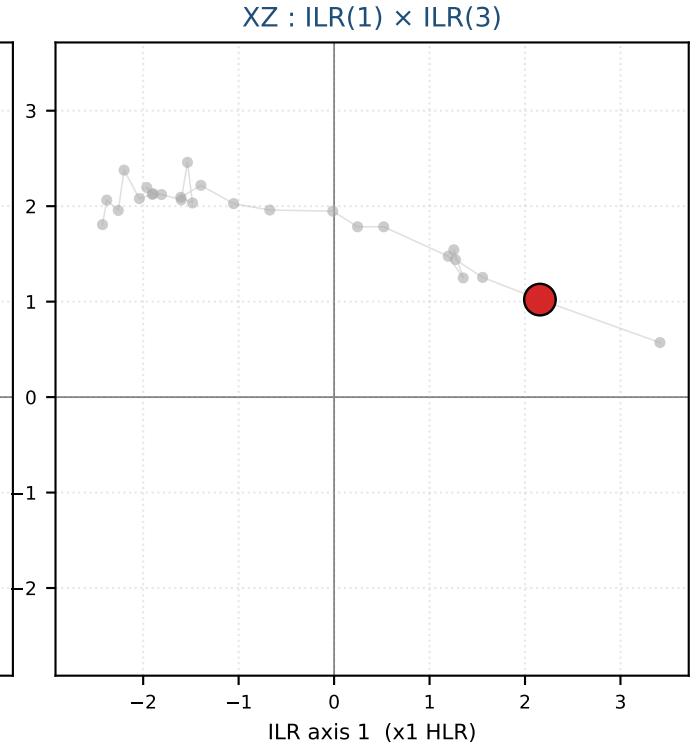
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2024 | D=9 T=26 | level 1 first-principles

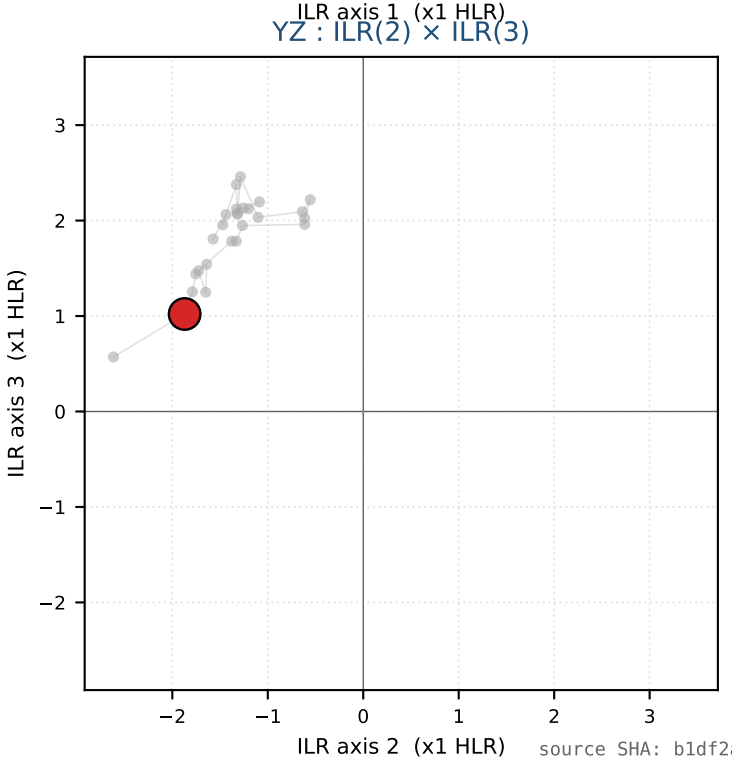
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



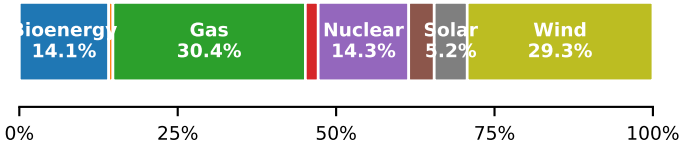


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1411	+4.884
Coal	0.0067	+1.835
Gas	0.3037	+5.651
Hydro	0.0203	+2.944
Nuclear	0.1428	+4.897
Other Fossil	0.0403	+3.631
Other Renewabl	0.0000	-33.346
Solar	0.0520	+3.887
Wind	0.2931	+5.615



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

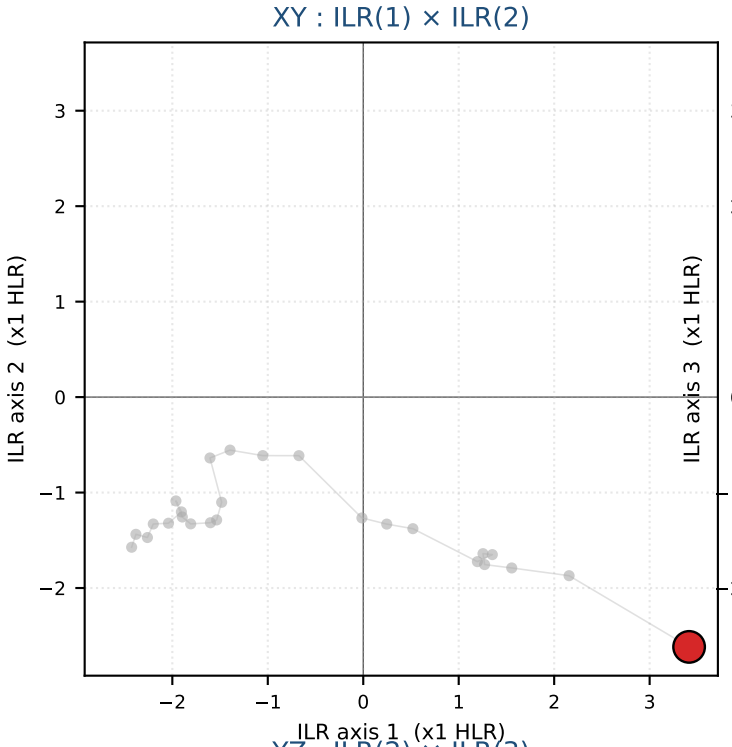
display factor: x1 (raw HLR x 1)

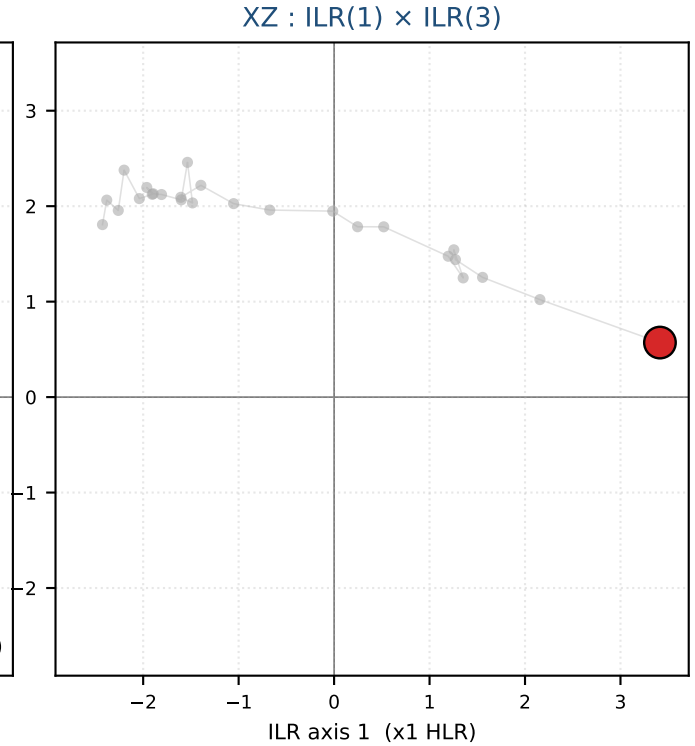
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=25 | 2025 | D=9 T=26 | level 1 first-principles

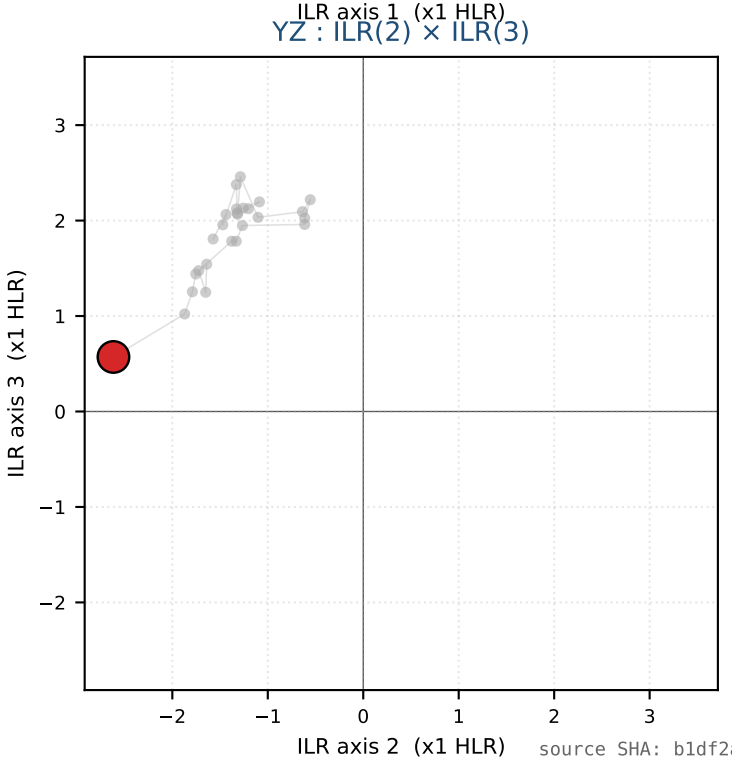
ember_gbr | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

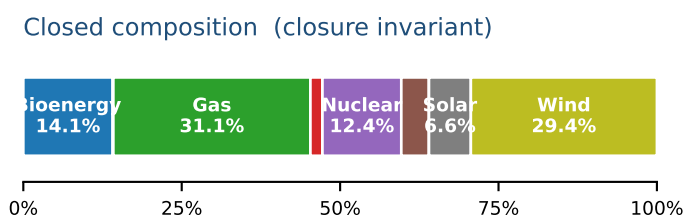




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.1410	+5.069
Coal	0.0011	+0.241
Gas	0.3110	+5.860
Hydro	0.0190	+3.064
Nuclear	0.1245	+4.944
Other Fossil	0.0437	+3.897
Other Renewabl	0.0000	-33.189
Solar	0.0661	+4.311
Wind	0.2937	+5.803





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

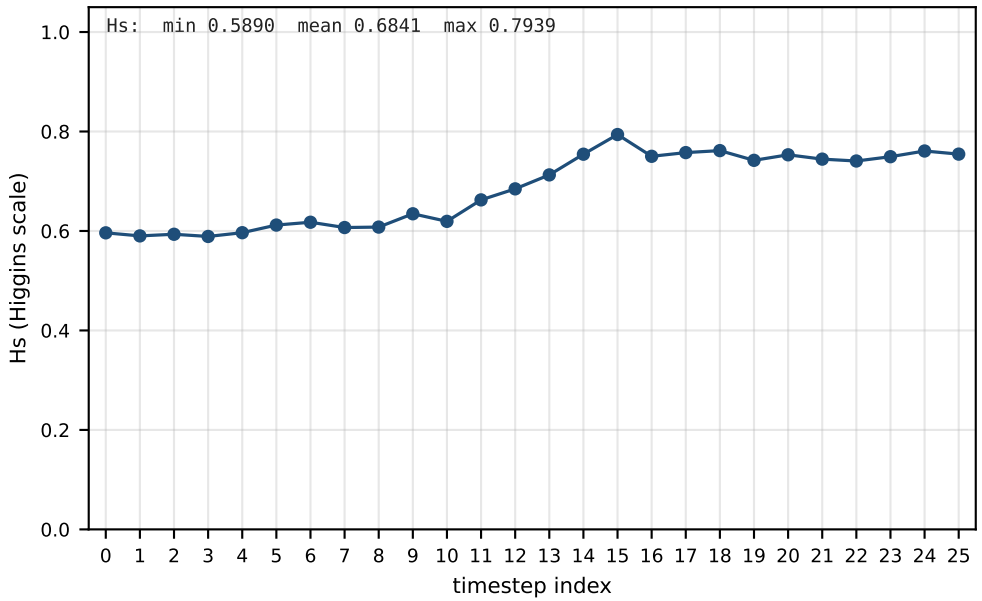
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

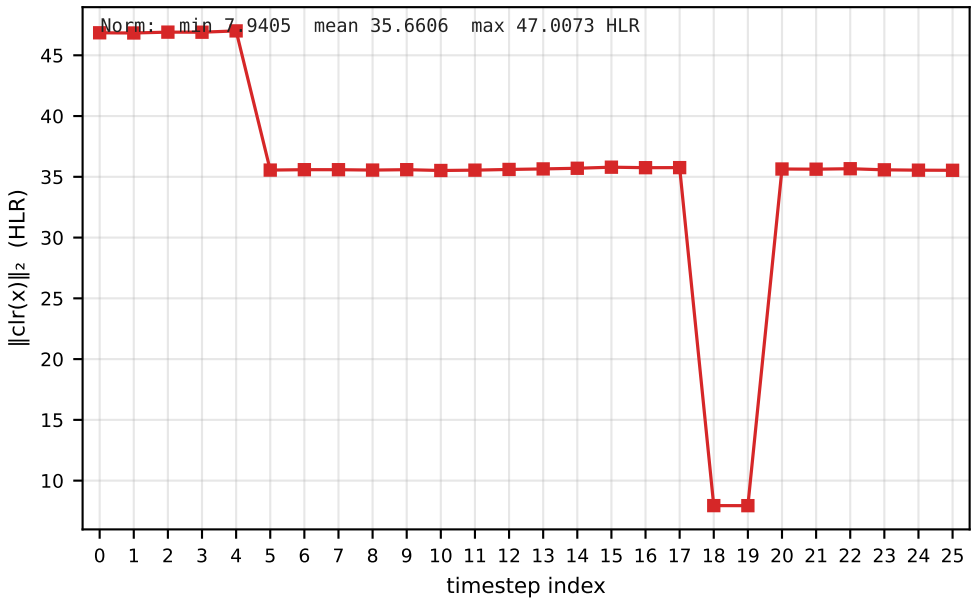
STAGE 1 SUMMARY | ember_gbr | D=9 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.3 | Schema 2.0.0 | Generated 2026-05-05T20:04:42

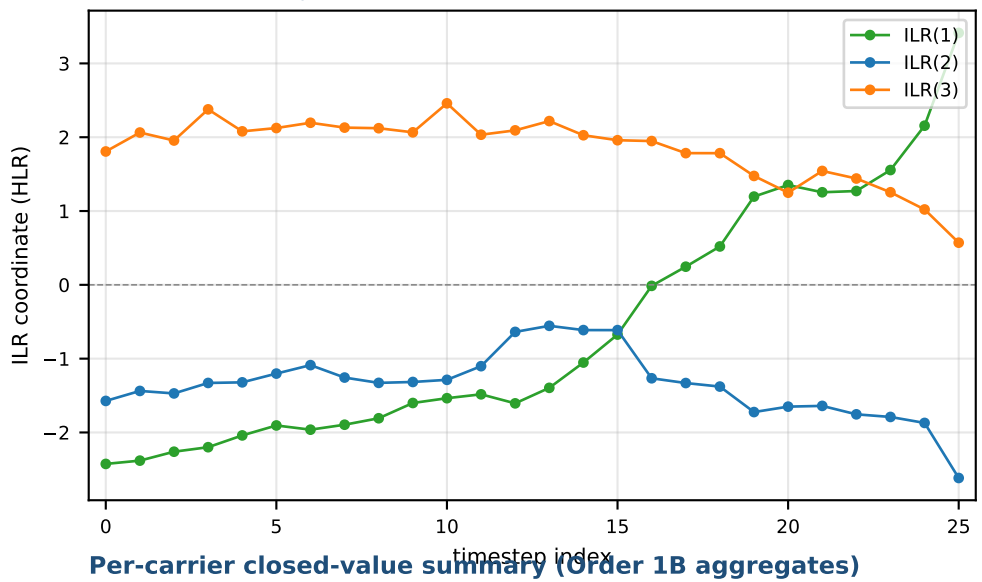
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 2.22e-16
closure status        :  OK ✓

Hs range              :  [0.5890, 0.7939]
Aitchison norm range  :  [7.9405, 47.0073] HLR
ILR(1) range          :  [-2.4263, +3.4134]
ILR(2) range          :  [-2.6168, -0.5552]
ILR(3) range          :  [+0.5711, +2.4591]

Source file           :  ember_GBR_United_Kingdom_generation_TWh.csv
Source SHA-256        :  b1df2a0ae6f21f77c0d6fb5b7a0c26af...
Closed-data SHA-256   :  e9e032d0675926fb170de906bf7bc053...
Content SHA-256       :  53e7b0c117c61f8a72beaa7f48dba330...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0103	0.0643	0.1411	0.0456
1	Coal	0.0011	0.2105	0.3924	0.1465
2	Gas	0.2675	0.3733	0.4597	0.0522
3	Hydro	0.0081	0.0151	0.0222	0.0034
4	Nuclear	0.1245	0.1835	0.2341	0.0308
5	Other Fossil	0.0233	0.0314	0.0437	0.0048
6	Other Renewabl	0.0000	0.0000	0.0000	0.0000
7	Solar	0.0000	0.0181	0.0661	0.0209
8	Wind	0.0025	0.1038	0.2937	0.1024